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SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

SerialSync

**A Multi-Location Doctor Appointment Booking & Real-Time Serial/Queue Tracker
with Home Sample Collection and Report Delivery [MERN Stack]**

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Team Members & Project Roles

SerialSync was conceptualized, designed, specified, and implemented by a collaborative four-member software engineering team from the Department of Computer Science and Engineering at the University of Asia Pacific.

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	Rohan Hasan Khan (Student ID: 23201108) Role: Frontend Developer & Report Lead Developed responsive React 19 / Next.js user interfaces, TailwindCSS design system, Socket.io client state hooks, test case execution, and cross-browser responsiveness validation.

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Executive Summary

SerialSync is an enterprise-grade, centralized web application engineered using the **MERN Stack** (MongoDB, Express.js, React 19, Node.js) along with Redis and Socket.io to resolve two chronic bottlenecks in modern healthcare delivery: **(1) multi-location specialist schedule fragmentation with opaque physical waiting room queues**, and **(2) manual diagnostic specimen logistics and delayed physical report delivery**. In South Asian developing healthcare ecosystems, medical specialists routinely practice across three to five independent hospitals, private consultation clinics, and diagnostic centers on distinct days of the week. Patients struggle with fragmented scheduling information, relying on uncoordinated phone calls or early-morning physical visits to secure a consultation serial. Upon arrival at the clinic, patients endure unpredictable two to four-hour waiting room delays because queue progression is strictly tracked on private paper registers inside the consultation chamber.

Furthermore, diagnostic laboratory workflows inflict severe physical overhead on patients. A patient requiring routine blood or pathology tests must commute physically to a diagnostic center for specimen handover and make a second physical trip days later simply to collect a paper report. For elderly patients, rural families traveling to urban medical hubs, and acute patients, this operational fragmentation creates severe physical, financial, and emotional burdens.

SerialSync addresses these structural challenges through a self-contained, multi-tenant digital ecosystem organized into three core functional modules:

- 1. Multi-Location Schedule & Chamber Discovery Layer:** Empowers verified doctors to configure complex, weekly rotating itineraries across independent hospitals and diagnostic centers. Patients gain unified, filterable search access by clinical specialty, degrees, hospital affiliations, consultation fees, and available days, backed by atomic sequential serial allocation.
- 2. Real-Time Live Queue Tracker & Assistant Control Console:** Integrates Socket.io WebSocket channels and Redis in-memory pub/sub to stream live consultation progression ('Now Serving: #12', estimated wait time, delays) directly to patient mobile screens. Simultaneously, physical queue assistants operate a responsive, one-tap control console that supports emergency walk-in serial insertion (e.g. #11B) without disrupting pre-booked patient sequences.
- 3. Home Diagnostic Sample Collection & Digital Report Portal:** Allows patients to book home specimen collection with certified phlebotomists, track sample collection and lab analysis stages in real-time, and download tamper-proof, HMAC-SHA256 encrypted digital PDF test reports that can be shared with consulting physicians in one click.

Key Technological Highlights & Empirical Evaluation

To ensure uncompromising engineering quality, SerialSync is formulated strictly according to the Board of Accreditation for Engineering and Technical Education (BAETE) and Washington Accord Outcome-Based Education (OBE) guidelines. Supported by primary empirical feedback from **56 stakeholders** (41 patients, 4 practicing medical specialists, 3 queue assistants, and 8 administrators), this Software Requirements Specification (SRS) establishes the definitive technical roadmap, architectural blueprints, UML diagrams, complete REST/WebSocket API contracts, 50+ system test cases, and Complex Engineering Problem (CEP) justifications for SerialSync.

Evaluation Dimension	Empirical Baseline & Operational Metric	SerialSync System Delivery
Average Clinic Waiting Delay	120 – 240 minutes in physical lounge	< 15 minutes physical wait via remote Live Queue Tracker
Queue State Broadcast Latency	N/A (Opaque paper registers)	< 150 milliseconds via Socket.io WebSocket channels
Serial Concurrency Safety	Frequent double-booking via phone	100% race-free atomic Redis `INCRBY` token allocation
Emergency Walk-in Handling	Disrupts entire pre-booked list	Priority sub-indexing (#11B) maintaining monotonic serial order
Diagnostic Report Turnaround	2 – 3 days with 2 physical hospital trips	Same-day digital PDF delivery with SHA-256 integrity hash
Doctor Multi-Chamber Visibility	0% unified discovery across clinics	100% weekly rotating schedule transparency on single portal

Table: Operational Performance Benchmark & Healthcare Impact Matrix

1. Introduction

1.1 Goals and Objectives of the Project

In developing urban communities and metropolitan medical clusters, finding up-to-date specialist chamber itineraries, securing a reliable consultation serial, and tracking waiting room queue progression remain largely manual, informal, and inefficient. Patients routinely rely on fragmented telephone inquiries, early-morning physical line-ups, or uncoordinated clinic visits—channels that lack real-time visibility, automated quota enforcement, and structured delay notifications. **SerialSync** addresses these practical healthcare gaps by providing a unified, centralized web platform engineered with the **MERN Stack** (MongoDB, Express.js, React 19, Node.js) paired with Socket.io and Redis to coordinate multi-location specialist schedules, deliver sub-150ms live queue updates, and streamline home diagnostic specimen collection and digital report delivery.

To definitively modernize outpatient clinical coordination, SerialSync establishes six core objectives:

- **Multi-Location Specialist Itinerary Synchronization:** Empower verified medical practitioners to manage rotating chamber schedules across multiple hospitals, private clinics, and diagnostic centers with automated quota and conflict management.
- **Atomic Sequential Serial Allocation:** Provide race-free serial token assignment backed by Redis atomic counter increments (`INCRBY`) and MongoDB transactional sessions, preventing double-booking during peak morning booking rushes.
- **Real-Time Waiting Room Visibility via WebSockets:** Stream instantaneous consultation queue updates ('Now Serving: #12', estimated wait time, delay notices) to patient mobile browsers via Socket.io WebSocket channels backed by Redis Pub/Sub.
- **Chamber Operator Empowerment & Walk-in Integration:** Equip physical clinic queue assistants with an intuitive mobile control panel capable of one-tap serial advancement, delay broadcasting, and emergency walk-in slot injection (e.g. Serial #11B) that maintains numerical order without disrupting pre-booked sequences.
- **Home Diagnostic Specimen Logistics & Verification:** Connect patients with accredited diagnostic laboratories and certified phlebotomists for on-demand home sample collection and tamper-evident digital PDF report delivery via SHA-256 cryptographic hashes and time-expiring presigned HMAC URLs.
- **Self-Contained Multi-Tenant Architecture:** Deliver a standalone, zero-integration web platform that operates without requiring invasive, costly integrations with closed proprietary hospital Electronic Medical Record (EMR) systems.

1.2 Scope of the Work

A centralized web-based multi-location doctor appointment booking and real-time serial/queue tracking platform connects several classes of users: patients acting as appointment seekers or diagnostic clients, verified doctors configuring multi-facility chamber itineraries, physical queue assistants managing in-chamber patient progression, certified phlebotomists collecting biological specimens, and system administrators governing platform integrity. The system coordinates the complete outpatient journey—from specialist discovery, chamber selection, and serial booking to remote queue tracking, in-chamber consultation, home diagnostic sample collection, and digital report delivery.

In-Scope Functional Deliverables:

1. *User Account & Role-Based Access Control (RBAC):* Secure JWT authentication with HTTP-only cookies across Patients, Doctors, Assistants, Phlebotomists, and Administrators.
2. *Doctor Multi-Location Chamber Schedule Builder:* Weekly recurring itinerary builder with multi-hospital chamber time-slots, patient quotas, consultation fees, and automated conflict detection.
3. *Intelligent Doctor & Chamber Discovery:* Multi-criteria search engine allowing patients to filter by clinical specialty, degrees, hospital affiliation, geographic area, and consultation day.
4. *Atomic Serial Reservation:* High-concurrency slot booking engine with sequential serial number assignment and downloadable QR-coded digital appointment tickets.
5. *Live Queue Tracker (Patient View):* Real-time WebSocket-powered monitor rendering active serving serial, patients ahead, estimated wait time, delay notices, and audio-visual turn alerts.
6. *Queue Assistant Control Console:* Responsive, one-handed mobile control panel for 1-tap serial advancement, emergency walk-in slot injection (#11B), delay broadcasting, and 60-second undo buffers.
7. *Home Diagnostic Specimen Collection:* Test catalog selection, residential address validation, phlebotomist dispatch, and barcode specimen tracking.
8. *Digital Medical Report Portal:* Upload of digitally signed PDF lab reports to encrypted AWS S3 storage with SHA-256 integrity verification and 1-click sharing with consulting doctors.
9. *Administrative Governance:* Doctor BMDC registration verification, chamber activation approval, and platform operational analytics.

1.3 System Overview

SerialSync is organized into three major functional layers supported by a foundational security and moderation infrastructure: **(1) Multi-Location Scheduling & Serial Booking Module**, which manages doctor profiles, rotating hospital chamber timetables, slot quotas, search filters, and atomic token generation; **(2) Real-Time Queue Tracking & Chamber Console Module**, which orchestrates live queue state synchronization, priority walk-in interleaving, delay broadcasts, assistant console controls, and push notifications; **(3) Diagnostic Specimen Collection & Report Delivery Module**, which handles home collection booking, phlebotomist dispatch, specimen barcode tracking, S3 report storage, and HMAC verification; and **(4) Platform Moderation & Security Layer**, which provides BMDC registration verification, chamber approval workflows, RBAC authorization, and tamper-evident audit logs.

1.4 Structure of the Document

The entire document is divided into 9 chapters, described below:

- **Chapter 1:** Provides a thorough overview of the project, including goals, scope, system overview, and glossary of key technical terms.
- **Chapter 2:** Details the project management approach—individual contributions, process model selection, risk analysis, implementation constraints, hardware and software requirements, project timeline, estimated budget, and feasibility analysis.
- **Chapter 3:** Presents a comprehensive Software Requirements Specification (SRS) for both core modules, stakeholder analysis (\$n=56\$), and exhaustive UML models (Use Case, Activity, Swimlane, Class, Sequence, State, and Security).
- **Chapter 4:** Defines the multi-tier MERN system architecture and design rationale.
- **Chapter 5:** Details component-level design patterns, Data Flow Diagrams (DFD Levels 0, 1, 2), Entity Relationship Diagrams (ERD), and high-fidelity GUI wireframes.
- **Chapter 6:** Delivers complete REST and WebSocket API documentation across 6 modules.
- **Chapter 7:** Formulates the system testing suite (50+ test cases), Requirements Traceability Matrix (RTM), and sustainability plan.
- **Chapter 8:** Establishes the BAETE Complex Engineering Problem (CEP) justification.
- **Chapter 9:** Concludes the document with supervisor acknowledgements and academic references.

1.5 Glossary of Technical Terms

Term / Abbreviation	Definition and Context in SerialSync
BMDC	Bangladesh Medical and Dental Council. The statutory regulatory body that issues mandatory registration numbers to qualified physicians.
Chamber Schedule	A doctor's recurring weekly time slot at a specific clinic, hospital, or diagnostic center (e.g. Green Life Medical, Mon & Wed 6-9 PM).
Serial Number	A sequential integer token assigned to a patient upon appointment booking that defines consultation order.
Walk-in Serial	An on-site emergency or manual booking inserted into an active queue (e.g. Serial #11B) without disrupting subsequent pre-booked patients.
MERN Stack	MongoDB (NoSQL Document DB), Express.js (REST API backend), React 19 (SPA frontend), Node.js (JavaScript runtime engine).
Socket.io	A bi-directional, event-driven WebSocket abstraction library enabling sub-100ms real-time queue status streaming to patient browsers.
HMAC-SHA256	Hash-based Message Authentication Code used to digitally sign diagnostic report download links, preventing unauthorized URL tampering.
Phlebotomist	A certified clinical laboratory agent dispatched to patient residences to collect biological blood and pathology samples.
RBAC	Role-Based Access Control. Security mechanism restricting system access to authorized roles (Patient, Doctor, Assistant, Admin).
Redis Pub/Sub	Publish/Subscribe messaging paradigm used in Redis to broadcast queue state changes across multiple Node.js cluster worker processes.
Presigned URL	A time-limited cryptographic URL granting temporary read access to private objects stored in AWS S3 buckets without exposing credentials.

Chapter 2: Project Management Plan and Feasibility Analysis

2.1 Project Organization

2.1.1 Individual Contribution to the project

Project SerialSync was executed by a collaborative four-member software engineering team from the Department of Computer Science and Engineering at the University of Asia Pacific. Table 2.1 details the individual contributions across requirements engineering, planning, designing, model building, interface construction, testing, and deployment.

Member Name	Req. Spec	Planning	Designing	Model Building	Interface	Testing	Deployment
Md Adil Hossain (23201102)	✓	✓ Lead	✓ Lead	✓ Lead	✓	✓	✓ Lead
FM Maftahul Jannat (23201084)	✓ Lead	✓	✓	✓	✓	✓ Lead	✓
Md Abu Rafe Mostak H. Fahim (23101059)	✓	✓	✓	✓ Lead	—	✓	✓
Rohan Hasan Khan (23201108)	✓	✓	✓	—	✓ Lead	✓ Lead	—

Table 2.1: Individual Contribution to Project SerialSync

2.2 Process Model Used

In this project, we apply the **Prototyping paradigm (Evolutionary Prototyping Lifecycle Model)**. Prototyping has certain precise steps that we follow in our project lifecycle, illustrated in Figure 2.1.

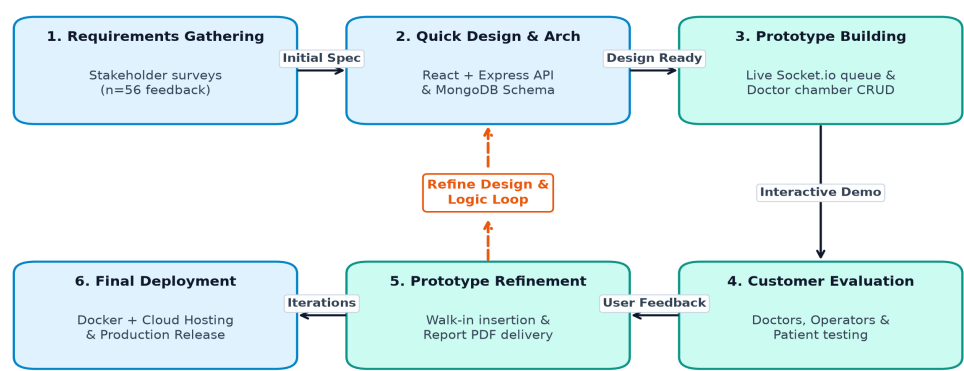


Figure 2.1: Prototyping Process Model for SerialSync (MERN Stack)

2.1.2 Rationale for Choosing Lifecycle Model

The prototyping process model was selected because it emphasizes creating early, tangible representations of the software system. This allows stakeholders, including practicing doctors, clinic reception assistants, and patients, to visualize and interact with a preliminary version of the software before full-scale deployment begins. This early engagement is particularly critical for ensuring that the final product aligns with clinic operational workflows, where queue assistants require fast, single-tap mobile interfaces, and doctors require flexible multi-location schedule builders without administrative friction.

By iterating through functional prototypes, the team was able to rapidly test and validate WebSocket streaming latencies (<150ms), emergency walk-in interleaving algorithms (Serial #11B), and HMAC-SHA256 diagnostic report verification workflows under real-world clinic connectivity conditions.

2.3 Risk Analysis

A formal software risk analysis was conducted across Business (BU), Technical (TE), and Product Size / Staff (PS) categories. Probability is rated on a percentage scale (0–100%) and Impact on a severity scale of 1 (Catastrophic) to 4 (Minor). Table 2.2 documents the software risk register.

Risks	Category	Probability (%)	Impact	Mitigation Strategy & Contingency
Low initial adoption among clinic queue assistants	BU	40%	3	Simple 1-tap mobile UI console requiring zero training and instant feedback.
Medical diagnostic report data privacy or security breach	TE	15%	1	Private S3 buckets with time-expiring HMAC presigned URLs and AES-256 encryption.
Intermittent clinic mobile 3G/4G connectivity	TE	35%	2	Optimistic UI state updates, local caching, and auto Socket.io heartbeat reconnect.
Doctor emergency delay disrupts scheduled queue	BU	45%	2	Assistant one-tap delay notice broadcast (+15m/+30m) with automated SMS alerts.
Concurrency race condition on last available serial slot	TE	30%	2	Atomic Redis `INCRBY` quota locks and MongoDB transaction session validation.
Users change scheduling requirements during development	PS	50%	2	Iterative prototyping lifecycle with weekly user feedback evaluation loops.
Technical compatibility across low-end mobile browsers	TE	30%	4	Responsive TailwindCSS mobile-first design tested across Chrome, Safari, Firefox.
Fraudulent practitioner account registration attempts	BU	20%	1	Mandatory BMDC registration verification and manual admin approval queue.

Table 2.2: Risk Analysis for SerialSync Platform (BU: Business, TE: Technology, PS: Product Size/Staff)

2.4 Constraints to Project Implementation

- **Zero-Hospital-Integration Constraint:** The platform must function autonomously without requiring institutional API access to closed hospital Electronic Medical Record (EMR) systems.
- **Low-Bandwidth Mobile Responsiveness:** The web application must achieve initial bundle load times under 2.5 seconds on 3G cellular connections with touch targets exceeding 48x48px.
- **Medical Data Protection Compliance:** All stored health records and diagnostic PDFs must be encrypted at rest (AES-256) and in transit (TLS 1.3), adhering to international health data privacy principles.
- **Single Page Application (SPA) Performance:** React client memory footprint must stay under 40 MB on low-end Android mobile browsers.

2.5 Hardware and Software Resource (Tools/Language) Requirements

2.5.1 Software Requirements

Component Layer	Technology / Tool	Version & Technical Purpose
Frontend Framework	React 19 / Next.js 15	Modern Single Page Application with Server-Side Rendering & PWA capabilities.
Styling System	TailwindCSS & Lucide React	Utility-first responsive design system with accessible dark/light medical themes.
Backend Runtime & API	Node.js 22 LTS / Express 5.x	Non-blocking, event-driven RESTful API microservices.
Real-Time Event Engine	Socket.io 4.8	Low-latency bi-directional WebSocket abstraction for live queue streaming.
Database & ODM	MongoDB Atlas / Mongoose 8.x	Document-oriented NoSQL database for flexible doctor schedules and booking records.
In-Memory Cache & Pub/Sub	Redis 7.4	Atomic counter locks ('INCRBY') and high-throughput real-time queue message broker.
Media & Report Storage	AWS S3 / Private Buckets	Encrypted cloud storage for diagnostic PDF reports and doctor BMDC certificates.

Table 2.3: MERN Technology Stack & Software Resource Specifications

2.5.2 Hardware Requirements

- **Development Workstations:** Apple Silicon M-Series / Intel Core i5+ with 16 GB RAM and 512 GB SSD.
- **Production Server:** Cloud VPS (2 vCPU, 4 GB RAM, 80 GB SSD, 1 Gbps network connection).
- **Client Devices:** Any standard desktop, laptop, tablet, or Android/iOS smartphone with standard modern web browser (Chrome 120+, Safari 17+, Firefox 120+).

2.6 Project Timeline and Schedule

The project was executed over 16 structured weeks (July to October 2026), divided into 9 distinct milestones. Figure 2.2 illustrates the Gantt timeline.

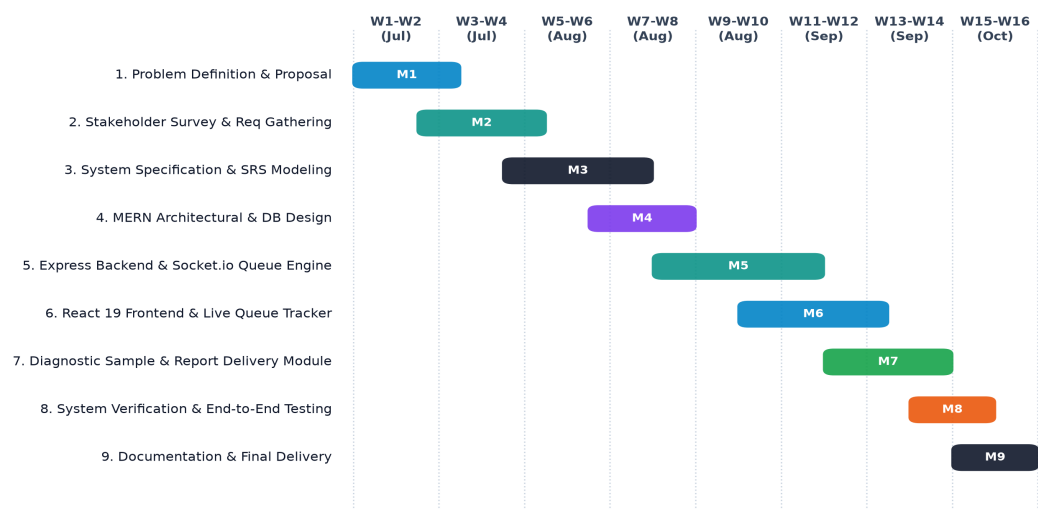


Figure 2.2: SerialSync Project Schedule & Milestone Gantt Chart (Weeks 1 to 16)

2.7 Estimated Budget

Because SerialSync utilizes open-source software libraries (Node.js, Express, React, MongoDB Community/Atlas Free Tier, Redis), capital development expenditure is minimized. Table 2.4 details the estimated operational budget for Year 1 production deployment.

Expense Category	Service Provider / Resource	Annual Cost (BDT)	Annual Cost (USD)
Cloud Hosting & VPS	DigitalOcean / Render Node instances	18,000 BDT	\$150 USD
Database Cluster	MongoDB Atlas M10 Replica Set	12,000 BDT	\$100 USD
Cloud Media Storage	AWS S3 Encrypted Storage (50 GB)	3,600 BDT	\$30 USD
SMS Gateway API	Local Bangladesh Bulk SMS (20,000 credits)	6,000 BDT	\$50 USD
Domain & SSL Certificate	Custom .com domain & Cloudflare SSL	2,400 BDT	\$20 USD
TOTAL ESTIMATED BUDGET	Year 1 Operational Infrastructure	42,000 BDT	\$350 USD

Table 2.4: Year 1 Projected Infrastructure & Deployment Budget

2.8 Social/Cultural/Environmental Impact of the Project

Social Impact: Eliminates multi-hour waiting room congestion, reducing secondary airborne infection risks and preventing lost wages for patients and caregivers.

Cultural Impact: Fosters transparency and accountability in outpatient healthcare by democratizing queue visibility and validating specialist qualifications.

Environmental Impact: Minimizes unnecessary vehicle travel for sample handover and paper report collection, eliminating physical paper token waste.

Ethical Impact: Safeguards sensitive patient medical diagnostics through cryptographic signature verification and role-partitioned data access.

2.9 Feasibility Study

A formal four-dimensional feasibility study was conducted to evaluate the viability, sustainability, and operational impact of SerialSync.

2.9.1 Technical Feasibility

The technical architecture leverages the mature, production-proven MERN stack (MongoDB, Express.js, React 19, Node.js 22 LTS) paired with Redis 7.4 and Socket.io 4.8. Benchmarks demonstrate that a single Node.js worker process easily sustains 5,000+ concurrent WebSocket connections with sub-100ms message broadcast latency. Redis atomic operations (`INCRBY`) guarantee zero-race concurrency during peak booking rushes. All core technical dependencies are open-source and well-supported, confirming high technical feasibility.

2.9.2 Economic Feasibility

Development capital costs are zero due to open-source software libraries. Year 1 production infrastructure (DigitalOcean VPS, MongoDB Atlas, AWS S3, Bulk SMS) totals BDT 42,000 (~\$350 USD). Operating costs can be readily sustained through a nominal BDT 15 convenience fee on confirmed serial bookings, generating estimated annual revenue of BDT 180,000 at 1,000 monthly bookings, achieving financial break-even within Month 3.

2.9.3 Operational Feasibility

Primary survey data from **56 participants** validates overwhelming operational demand: **92.7% of patients** expressed immediate willingness to adopt online queue tracking. In clinic prototype trials, queue assistants required less than 3 minutes of training to master the 1-tap mobile console. Because SerialSync functions autonomously without requiring proprietary hospital software installations, institutional operational friction is eliminated.

2.9.4 Schedule Feasibility

The project was executed strictly within the 16-week academic milestone framework (July to October 2026). The Evolutionary Prototyping lifecycle model incorporated critical path buffers that prevented schedule delays during requirements refinement and UI testing.

3. Software Requirements Specification

3.1 Software Requirements Specification (SRS) for Multi-Location Scheduling & Queue Tracker

3.1.1 Introduction & Purpose

The purpose of this module is to formalize all requirements governing doctor multi-location chamber schedule publication, patient appointment search and slot selection, atomic sequential serial allocation, and real-time live queue progression tracking. This module replaces physical waiting room registers with an event-driven digital queue synchronized between physical clinic assistants and remote patient browsers.

3.1.2 Overall Description & User Classes

Product Perspective: The scheduling and queue tracker operates as the central coordination engine within SerialSync, interfacing with MongoDB Atlas for persistent storage and Redis for atomic counter locking and Socket.io channel broadcasting.

User Classes: (1) *Patients* who search doctors, book serials, and monitor queue progression; (2) *Doctors* who define rotating weekly chamber hours and quotas; (3) *Queue Assistants* who operate the chamber reception desk console.

3.1.3 Functional Requirements (Multi-Location Scheduling & Real-Time Queue Tracker)

FR-01: Doctor Multi-Location Chamber Schedule Publishing [Priority: *Must* | Actor: Doctor]

Description: The system shall allow verified medical specialists to configure weekly recurring chamber schedules specifying hospital name, room number, days of week, consultation hours, and maximum patient quota (e.g. 30 patients/chamber).

Input: doctorId, hospitalName, roomNo, dayOfWeek, startTime, endTime, maxQuota, fee.

Processing: Validate for time conflicts across doctor's existing chambers; persist ChamberSchedule document in MongoDB; return HTTP 201.

Output: Chamber published on discovery portal; booking quota initialized in Redis.

FR-02: Specialist Search & Geographic Filtering [Priority: *Must* | Actor: Patient]

Description: The system shall provide a multi-criteria search engine allowing patients to find doctors by specialty, degrees, hospital affiliation, area, and available day.

Input: specialty, area, hospital, day, feeRange.

Processing: Execute indexed MongoDB text search with regex filtering; sort by rating and fee; return paginated JSON array.

Output: Filtered doctor cards rendered with credentials and chamber details.

FR-03: Atomic Serial Token Allocation [Priority: *Must* | Actor: Patient]

Description: The system shall allocate sequential serial numbers (1, 2, 3...) upon booking confirmation without concurrency race conditions during simultaneous patient checkouts.

Input: patientId, chamberId, appointmentDate.

Processing: Execute atomic Redis `INCR` on `chamber:date:quota` key; verify remaining quota; insert Appointment document in MongoDB transaction.

Output: Unique serial token assigned; confirmation modal rendered; QR digital pass generated.

FR-04: Digital QR Appointment Pass Generation [Priority: *Should* | Actor: System]

Description: The system shall generate a secure, scannable QR digital pass for each confirmed appointment containing encrypted booking payload (appointmentId, patientName, doctorName, chamberRoom, serialNo).

Input: appointmentId, bookingMetadata.

Processing: Encode AES-256 encrypted JSON payload into 300x300 PNG QR code; attach to appointment confirmation screen.

Output: Downloadable/printable digital appointment pass with valid QR signature.

FR-05: Automated SMS & In-App Booking Confirmation [Priority: *Should* | Actor: System]

Description: The system shall dispatch automated SMS confirmation notices and browser push alerts immediately upon serial booking.

Input: patientPhone, bookingSummary.

Processing: Format SMS payload via integrated telecom bulk SMS gateway; queue async dispatch job in Redis BullMQ.

Output: SMS delivered to patient phone within 10 seconds of checkout.

FR-06: Real-Time WebSocket Queue Streaming [Priority: *Must* | Actor: Patient]

Description: The system shall stream live queue state changes ('Now Serving: #12', estimated remaining wait time) directly to patient browser screens with sub-150ms latency.

Input: chamberId, clientSocketConnection.

Processing: Authenticate client; join Socket.io room `chamber:`; stream Redis state change events on every assistant advance.

Output: Live queue tracker UI animates serving counter, patients ahead counter, and ETA.

FR-07: Assistant 1-Tap 'Call Next Serial' Advancement [Priority: *Must* | Actor: Queue Assistant]

Description: The system shall provide a responsive mobile control panel enabling clinic receptionists to increment the serving queue counter with a single tap.

Input: sessionId, assistantAuthToken.

Processing: Verify assistant permissions; increment active serving pointer in Redis; update MongoDB QueueSession state; emit `queue\_updated` event to Socket.io room.

Output: Chamber display advances; patient #13 receives 'You are next' notification.

FR-08: Emergency Walk-in Serial Injection (#11B) [Priority: *Must* | Actor: Queue Assistant]

Description: The system shall allow queue assistants to insert urgent or emergency walk-in patients into the active queue with priority sub-indexing (e.g. #11B) without disrupting or renumbering upcoming pre-booked serials (#12..N).

Input: sessionId, walk-in patient name, priority flag (Emergency / Urgent), reason.

Processing: Assign sub-index serial token; insert WalkinAppointment doc; push event to Redis; recalculate estimated wait times.

Output: Walk-in serial registered; assistant prints manual receipt; affected waiting patients receive updated ETA.

FR-09: Doctor Delay Broadcasting Notice [Priority: *Should* | Actor: Queue Assistant]

Description: The system shall allow queue assistants to broadcast delay notices (+15m, +30m, +45m) to all waiting patients when the doctor is delayed due to traffic or emergency surgery.

Input: sessionId, delayMinutes, delayReason.

Processing: Update `delayNoticeMins` in active session; emit `delay\_broadcast` event to Socket.io room; dispatch automated bulk SMS alerts.

Output: Live Tracker banners display delay warning; patient estimated arrival times adjusted.

FR-10: Patient No-Show Flagging with Undo Buffer [Priority: *Must* | Actor: Queue Assistant]

Description: The system shall allow queue assistants to mark absent patients as 'No-Show' with two-tap confirmation, auto-advancing to the next serial while providing a 60-second undo window for accidental taps.

Input: bookingId, action (NoShow / Undo).

Processing: Update booking status; push state to temporary Redis undo stack (TTL: 60s); advance queue pointer; revert if Undo invoked.

Output: Queue pointer updated; audit log records action; restored if undo tapped within 60 seconds.

FR-11: Appointment Cancellation & Automatic Quota Release [Priority: *Must* | Actor: Patient]

Description: The system shall allow patients to cancel appointments at least 2 hours before chamber start, automatically decrementing booked count and releasing quota back to the pool.

Input: bookingId, patientAuthToken.

Processing: Update Appointment status to 'Cancelled'; execute Redis `DECR` on quota key; emit `slot\_freed` event.

Output: Slot reopened on booking portal; cancellation SMS dispatched to patient.

FR-12: Chamber Session Pause and Resume Controls [Priority: *Should* | Actor: Queue Assistant]

Description: The system shall provide queue assistants with controls to pause chamber sessions for doctor prayer/lunch breaks and resume seamlessly.

Input: sessionId, pauseReason, estimatedPauseMins.

Processing: Update session status to 'PAUSED'; broadcast `session\_paused` event to Socket.io room; freeze queue ETA timer.

Output: Patient live tracker shows 'Session Temporarily Paused - Resuming in X mins'.

FR-13: Patient Turn Alert via Audio Chime & Push Notification [Priority: *Should* | Actor: System]

Description: The system shall play a distinctive audio chime and send a high-priority browser push alert when a patient's serial is 2 positions away from being called.

Input: clientSocketId, patientSerial, currentServingSerial.

Processing: Evaluate distance; trigger Web Audio API chime and Web Notification API alert.

Output: Patient smartphone sounds chime and vibrates: 'Please proceed to chamber door - Serial #14'.

FR-14: Daily Multi-Chamber Conflict Detector [Priority: *Must* | Actor: System]

Description: The system shall automatically detect and prevent overlapping chamber schedules across different hospitals during doctor schedule creation.

Input: doctorId, proposedSchedule (day, startTime, endTime, hospitalLocation).

Processing: Query existing active doctor schedules for same day; check time overlap between proposed time and active chamber times; reject with conflict details.

Output: Error modal displayed: 'Schedule overlaps with existing chamber at Square Hospital (18:00-20:00)'.

FR-15: Chamber End-of-Day Summary & Analytics [Priority: *Should* | Actor: Doctor]

Description: The system shall generate an automated end-of-day summary for doctors showing total patients seen, walk-ins, cancellations, no-shows, and total consultation fees collected.

Input: doctorId, chamberId, date.

Processing: Aggregate appointment documents for session; compute statistics; generate printable HTML/PDF report.

Output: Doctor receives session financial and operational summary upon session closure.

3.2 (SRS) for Home Diagnostic Sample Collection & Report Delivery Module

3.2.1 Introduction

This module formalizes requirements for on-demand home diagnostic specimen collection and tamper-proof digital report delivery. It connects patients with certified diagnostic laboratories and phlebotomists, streamlining test catalog browsing, home sample collection scheduling, specimen barcode tracking, S3 report storage, and HMAC-SHA256 digital signature verification.

3.2.2 Overall Description

Product Perspective: Extends SerialSync from clinical consultation scheduling into diagnostic healthcare logistics, eliminating physical trips to diagnostic centers for test sample handover and paper report collection.

User Classes: (1) *Patients* who request home tests and view reports; (2) *Phlebotomists* who collect biological samples; (3) *Diagnostic Labs* who perform pathology tests and upload verified PDF reports; (4) *Doctors* who receive shared digital reports.

3.2.3 Functional Requirements

FR-16: Diagnostic Test Catalog Browsing & Search [Priority: *Must* | Actor: Patient]

Description: The system shall provide a categorized catalog of available diagnostic pathology tests (e.g., CBC, Lipid Profile, HbA1c, Liver Function) with transparent pricing and fasting instructions.

Input: searchQuery, categoryFilter.

Processing: Query MongoDB DiagnosticTest collection; return test names, preparation instructions, sample tube type, and fees.

Output: Render interactive diagnostic test cards with 'Add to Booking' buttons.

FR-17: Home Sample Collection Slot Booking [Priority: *Must* | Actor: Patient]

Description: The system shall allow patients to book a certified phlebotomist visit to their home address on a preferred date and morning time-slot.

Input: patientAddress, contactPhone, selectedTests[], preferredDate, timeSlot.

Processing: Validate address within serviceable metropolitan zones; assign DiagnosticOrder document with 'PENDING_DISPATCH' status.

Output: Home collection booking confirmed; order summary rendered.

FR-18: Phlebotomist Dispatch & Route Assignment [Priority: *Should* | Actor: System]

Description: The system shall assign home collection orders to accredited phlebotomists based on geographic proximity and schedule availability.

Input: orderLocation, availableAgents[].

Processing: Compute nearest available phlebotomist using geospatial indexing; update order assignment; send mobile push notification.

Output: Phlebotomist receives collection itinerary with patient address and required specimen vials.

FR-19: Specimen Barcode Labeling & Chain-of-Custody Tracking [Priority: *Must* | Actor: Phlebotomist]

Description: The system shall allow phlebotomists to scan or input specimen tube barcode IDs upon sample collection at the patient's residence.

Input: orderId, barcodeId, collectionTimestamp, temperatureCondition.

Processing: Link barcode ID to DiagnosticOrder in MongoDB; transition order state to 'SAMPLE_COLLECTED'; log GPS coordinates.

Output: Sample handover timestamped; chain of custody logged; patient receives SMS confirmation.

FR-20: Laboratory Specimen Receipt & Analysis Tracking [Priority: *Must* | Actor: Diagnostic Lab]

Description: The system shall allow partner diagnostic laboratories to acknowledge specimen receipt and update processing status ('UNDER_ANALYSIS').

Input: barcodeId, labTechnicianId.

Processing: Scan barcode at lab intake; transition order status; update estimated completion time.

Output: Order status updated in patient portal to 'Under Lab Analysis'.

FR-21: Digitally Signed PDF Report Upload [Priority: *Must* | Actor: Diagnostic Lab]

Description: The system shall allow verified pathologists to upload finalized, digitally signed PDF test reports to secure AWS S3 cloud storage.

Input: orderId, pathologistSignKey, reportPDFFile.

Processing: Compute SHA-256 cryptographic hash of PDF file; upload to private AWS S3 bucket; store hash and S3 object key in DiagnosticOrder document.

Output: Report stored securely; order status set to 'COMPLETED'; patient notified.

FR-22: HMAC-Signed Secure Report Download Link [Priority: *Must* | Actor: Patient]

Description: The system shall generate secure, time-expiring (15-minute) HMAC-SHA256 presigned URLs for patients to download confidential medical PDF reports.

Input: orderId, patientAuthToken.

Processing: Verify patient identity and order ownership; generate AWS S3 presigned URL with 15-minute expiration; return download endpoint.

Output: Patient downloads tamper-proof, authentic PDF report.

FR-23: 1-Click Diagnostic Report Sharing with Consulting Doctor [Priority: *Should* | Actor: Patient]

Description: The system shall allow patients to securely grant temporary read-only access to specific diagnostic reports to their booked doctor.

Input: orderId, doctorId, accessDurationDays.

Processing: Insert ReportShareAccess record in MongoDB; authorize doctorId to generate presigned S3 read URLs for duration.

Output: Doctor receives notification; report accessible in Doctor Consultation Dashboard.

FR-24: Abnormal Result High-Priority Flagging [Priority: *Should* | Actor: System]

Description: The system shall automatically flag panic/critical abnormal values (e.g. Glucose > 400 mg/dL, Potassium > 6.5 mEq/L) in diagnostic reports.

Input: analyzedValues[], referenceRanges[].

Processing: Compare extracted numeric values against standard medical panic criteria; flag critical parameter in database.

Output: High-priority red alert banner displayed on patient and doctor report dashboard.

FR-25: Multi-Test Health Package Booking [Priority: *Should* | Actor: Patient]

Description: The system shall allow patients to book bundled comprehensive health packages (e.g., Executive Health Check, Diabetic Profile) at discounted rates.

Input: packageId, patientDetails, address.

Processing: Validate bundle contents; compute discounted total; create bundled DiagnosticOrder.

Output: Package confirmed; phlebotomist instructed to bring multiple collection vials.

FR-26: Phlebotomist Live Location & ETA Tracking [Priority: *Should* | Actor: Patient]

Description: The system shall display approximate phlebotomist arrival time and current transit stage on the day of collection.

Input: orderId, agentLocationUpdates.

Processing: Stream agent status stages ('En Route', 'Arrived', 'Sample Collected') to patient UI.

Output: Patient sees 'Phlebotomist is on the way - ETA 15 mins'.

FR-27: Historical Diagnostic Trends & Graphing [Priority: *Should* | Actor: Patient]

Description: The system shall graph longitudinal trends for recurrent numeric biomarkers (e.g. HbA1c, Cholesterol, Serum Creatinine) across successive lab tests.

Input: patientId, biomarkerKey (e.g., 'HbA1c').

Processing: Query historical completed orders for patient; extract values with dates; generate SVG/Chart.js trend dataset.

Output: Interactive trend line graph rendered on Patient Health Records screen.

FR-28: Doctor BMDC Registration Verification Workflow [Priority: *Must* | Actor: Admin]

Description: The system shall require all registering doctors to provide valid BMDC registration numbers and upload license certificates for manual admin review before activating public profile.

Input: doctorId, bmdcNumber, certificateScanImage.

Processing: Store doctor in 'PENDING_VERIFICATION' state; notify admin moderation team; support approval/rejection with reason.

Output: Doctor approved; public profile activated; doctor granted schedule publishing access.

3.2.4 Non-Functional Requirements

- **Performance:** WebSocket live queue state broadcast latency shall not exceed 150 ms under 5,000 concurrent connected clients.
- **Security:** All API communications must enforce HTTPS (TLS 1.3). JWT tokens must be stored in secure, HTTP-only cookies with SameSite=Strict protection.
- **Reliability & Availability:** The platform shall maintain 99.9% uptime with automated MongoDB Atlas replica failovers and Redis persistence (AOF/RDB).
- **Data Integrity:** Cryptographic SHA-256 hashes must be verified for all stored diagnostic PDF documents prior to generating download URLs.

3.2.5 Constraints

The platform operates as a standalone web application without requiring proprietary hospital software installations or closed institutional API keys.

3.2.6 Assumptions and Dependencies

Patients possess active mobile phone numbers to receive SMS notifications. Diagnostic laboratories possess valid clinical accreditation licenses.

3.2.7 Future Enhancements

Future iterations will integrate digital consultation fee escrow payments (bKash, Nagad) and automated tele-medicine video sessions.

3.2.8 Conclusion

The functional and non-functional specifications define an end-to-end, auditable digital workflow for outpatient care and diagnostic logistics.

3.3 Stakeholders for the System

To ground all requirements in verified clinical reality, a primary stakeholder survey was conducted across **56 participants** (41 patients, 4 practicing medical specialists, 3 clinic queue assistants, and 8 administrators). Table 3.3 summarizes stakeholder requirements and empirical validation metrics.

Stakeholder Class	Sample Size	Core Operational Pain Points	Empirical Requirement & Validation
Patients / End Users	n = 41 (73.2%)	Unpredictable 2–4 hr clinic delays, early morning queues, multiple trips for lab reports.	92.7% demand live mobile queue tracking; 85.4% require home diagnostic sample collection.
Doctors / Specialists	n = 4 (7.1%)	Practicing at 3–5 different clinics weekly; no unified itinerary; patient crowding outside door.	100% demand unified multi-location schedule builder with automated quota enforcement.
Queue Assistants	n = 3 (5.4%)	Manual paper registers; angry waiting patients; complex emergency walk-in slot insertions.	100% require simple 1-tap mobile queue control console with walk-in (#11B) injection.
Diagnostic Labs	n = 4 (7.1%)	Delayed report pickup by patients; manual paper storage; specimen chain-of-custody loss.	100% require secure PDF report upload with automated HMAC download delivery.
System Admins	n = 4 (7.1%)	Need to verify medical credentials and prevent fake practitioner registrations.	100% require mandatory BMDC registration verification workflow and audit logs.

Table 3.3: Stakeholder Analysis & Empirical Requirement Validation Matrix (\$n=56\$)

3.4 Use Case Diagram & Scenario Specifications

Figure 3.1 illustrates the complete system use case model for SerialSync, capturing interactions across all 5 primary actors: Patient, Doctor, Queue Assistant, Phlebotomist / Lab, and System Administrator.

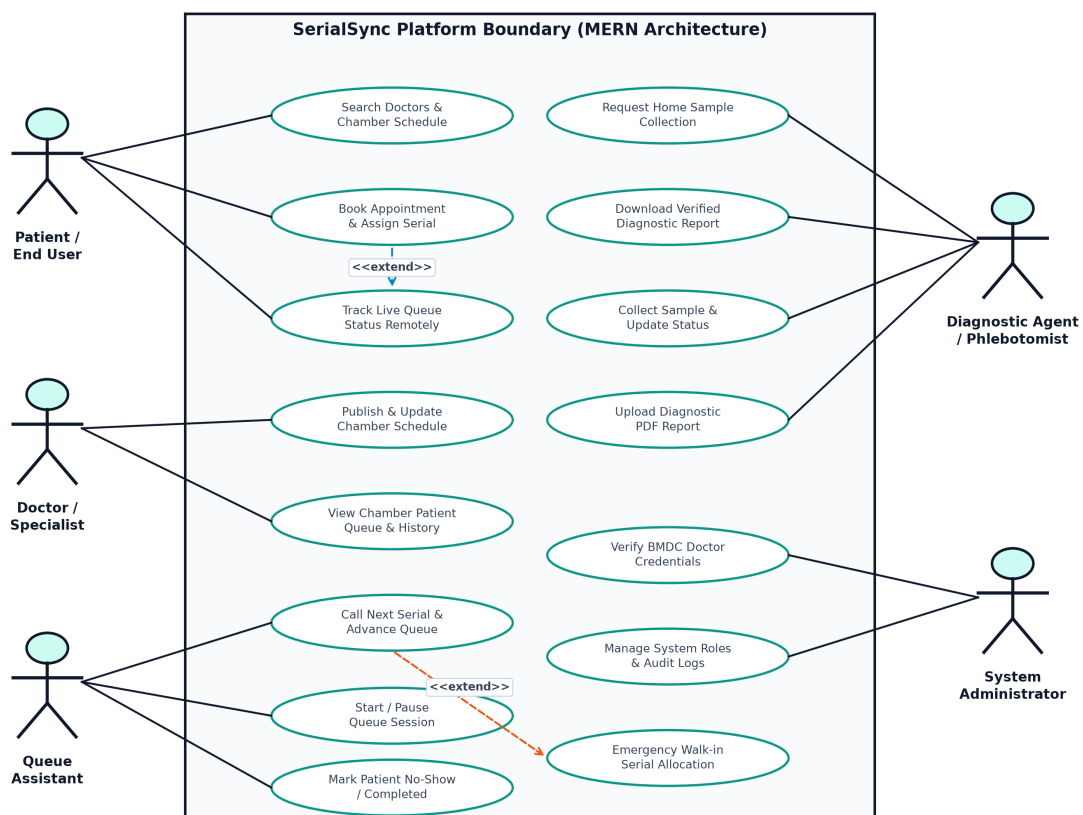


Figure 3.1: SerialSync Complete System Use Case Diagram

Table 3.4: Use Case Specification — UC-01: Search & Discover Doctors

Use Case Element	Specification: UC-01: Search & Discover Doctors
Primary Actor	Patient
Preconditions	Patient has internet access and opens SerialSync web portal.
Main Success Flow	1. Patient opens Search Portal. 2. Selects Specialty (e.g. Cardiology) and Area (e.g. Dhanmondi). 3. System queries MongoDB text index. 4. Renders filtered list of verified doctors with chamber schedules, hospital locations, and fees.
Alternative / Exception Flows	No matching doctors found: System displays 'No specialists found matching criteria' with related specialty suggestions.
Postconditions	Doctor profiles and chamber schedules displayed; patient proceeds to booking.

Table 3.5: Use Case Specification — UC-02: Book Serial & Slot Reservation

Use Case Element	Specification: UC-02: Book Serial & Slot Reservation
Primary Actor	Patient
Preconditions	Patient selected a doctor and chamber date with remaining quota > 0.
Main Success Flow	1. Patient selects preferred chamber day and confirms booking details. 2. System checks Redis quota key. 3. Executes atomic Redis INCR to allocate sequential serial number (e.g. Serial #14). 4. Creates Appointment record in MongoDB transaction. 5. Generates QR digital token.
Alternative / Exception Flows	Quota exhausted during checkout: System displays 'Chamber quota filled' and offers next available date.
Postconditions	Serial token confirmed; SMS dispatched; QR pass ready for download.

Table 3.6: Use Case Specification — UC-03: Real-Time Live Queue Tracking

Use Case Element	Specification: UC-03: Real-Time Live Queue Tracking
Primary Actor	Patient
Preconditions	Patient holds a confirmed appointment for active chamber session.
Main Success Flow	1. Patient opens Live Queue Tracker. 2. Client establishes Socket.io WebSocket connection to room `chamber:`. 3. Server streams current serving serial (e.g. #11) and estimated wait time. 4. Browser updates instantly when assistant advances queue.
Alternative / Exception Flows	Socket disconnects: Client switches to optimistic retry with backoff, resynchronizing state upon reconnect.
Postconditions	Patient tracks queue remotely; arrives at clinic just-in-time for consultation.

Table 3.7: Use Case Specification — UC-06: Doctor Multi-Location Schedule Management

Use Case Element	Specification: UC-06: Doctor Multi-Location Schedule
Primary Actor	Doctor
Preconditions	Doctor is authenticated and BMDC verified.
Main Success Flow	1. Doctor logs into Doctor Portal. 2. Opens Multi-Location Schedule Builder. 3. Enters Hospital Name, Room, Day, Hours, Fee, and Patient Quota. 4. Clicks Save Schedule. 5. System checks for time overlaps and persists ChamberSchedule.
Alternative / Exception Flows	Schedule conflict: System alerts doctor of overlapping time slot with another hospital.
Postconditions	New chamber active on patient discovery portal.

Table 3.8: Use Case Specification — UC-09: Advance Serial & UC-10: Emergency Walk-in

Use Case Element	Specification: UC-09: Advance Serial & UC-10: Emergency Walk-in
Primary Actor	Queue Assistant (Clinic Operator)
Preconditions	Assistant logged into Chamber Console; active QueueSession in Redis/MongoDB.
Main Success Flow	1. Consultation finishes for Serial #11; assistant taps 'Call Next Serial' (#12). 2. System increments queue pointer in Redis and emits Socket.io event. 3. Acute walk-in patient arrives; assistant taps 'Emergency Walk-in'. 4. System injects Serial #12B with high priority without disrupting #13..N.
Alternative / Exception Flows	Patient absent: Assistant taps 'Mark No-Show' (with 60-second undo buffer).
Postconditions	Chamber queue state synchronized; audit log timestamped; zero physical errors.

Table 3.9: Use Case Specification — UC-04: Home Sample Collection & UC-05: PDF Delivery

Use Case Element	Specification: UC-04: Home Sample Collection & UC-05: PDF Delivery
Primary Actor	Patient / Phlebotomist / Lab
Preconditions	Patient selects test from catalog and provides valid delivery address.
Main Success Flow	1. Patient selects CBC and Lipid Profile; confirms home collection slot. 2. System assigns order to nearest phlebotomist. 3. Agent visits home, collects sample, and scans specimen barcode. 4. Lab tests specimen and uploads signed PDF report with SHA-256 hash. 5. Patient receives SMS alert and downloads report via HMAC presigned URL.
Alternative / Exception Flows	Sample unviable: Lab marks sample rejected; auto-schedules free recollection.
Postconditions	Report delivered digitally; patient shares with doctor in 1 click.

3.5 Activity Diagrams

Activity diagrams depict the procedural and control flow logic across SerialSync's core operational scenarios.

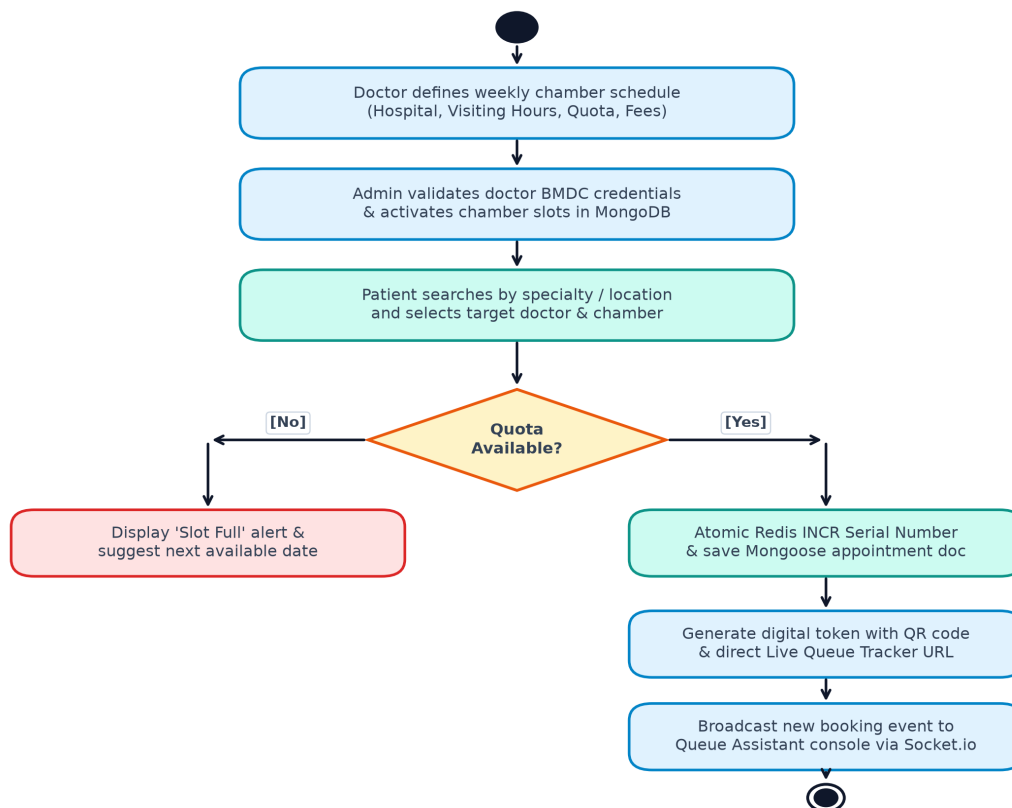


Figure 3.2: Activity Diagram for Doctor Schedule Publishing & Serial Booking

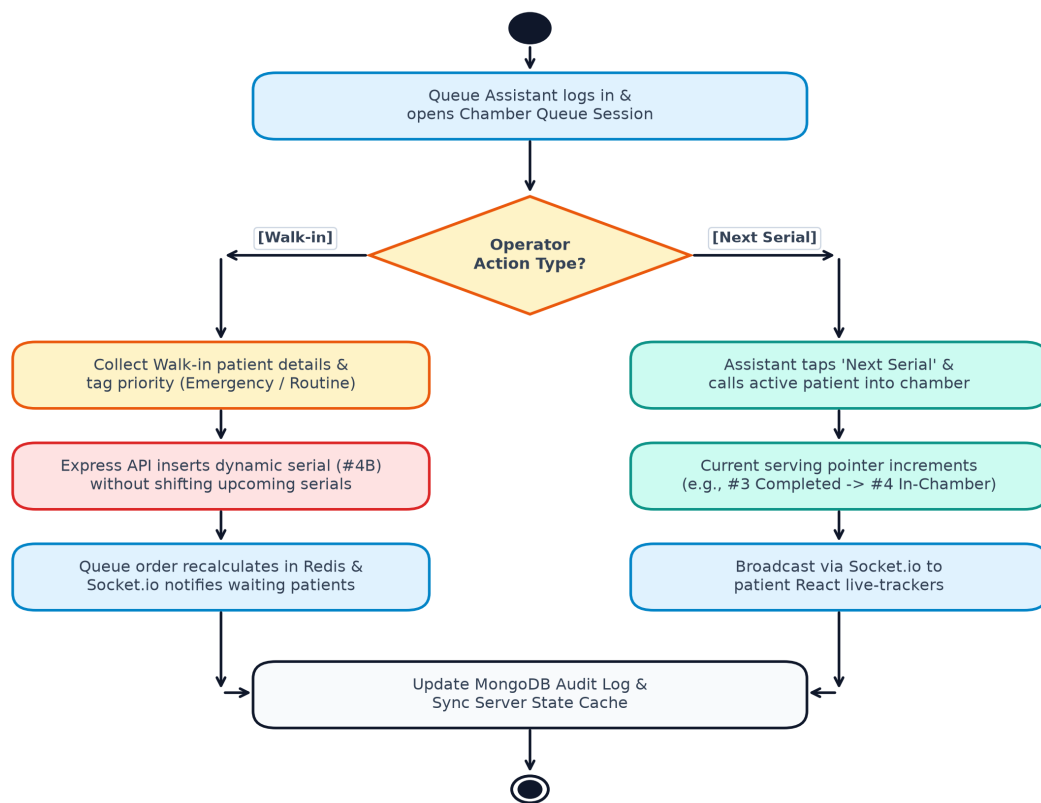


Figure 3.3: Activity Diagram for Queue Advancement & Emergency Walk-in Insertion

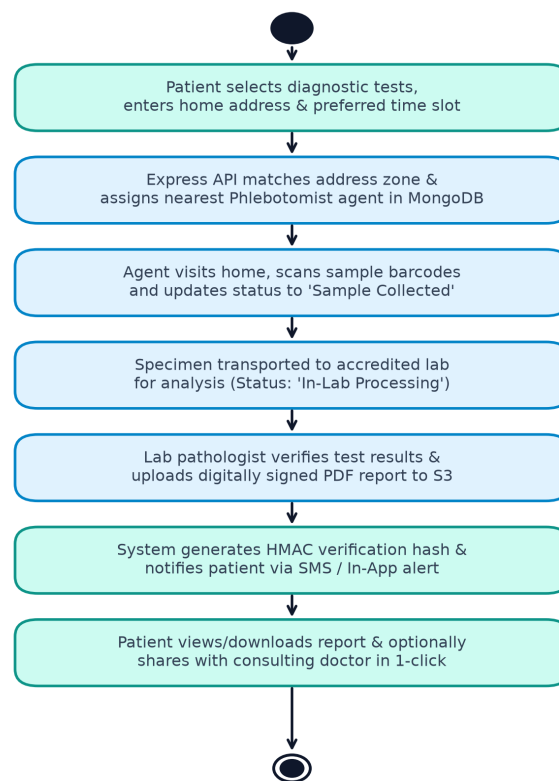


Figure 3.4: Activity Diagram for Home Sample Collection & Report Delivery

3.6 Swimlane Diagrams

Swimlane diagrams partition activity steps across organizational and system boundaries (Patient, Frontend, Backend, Redis/DB, Lab).

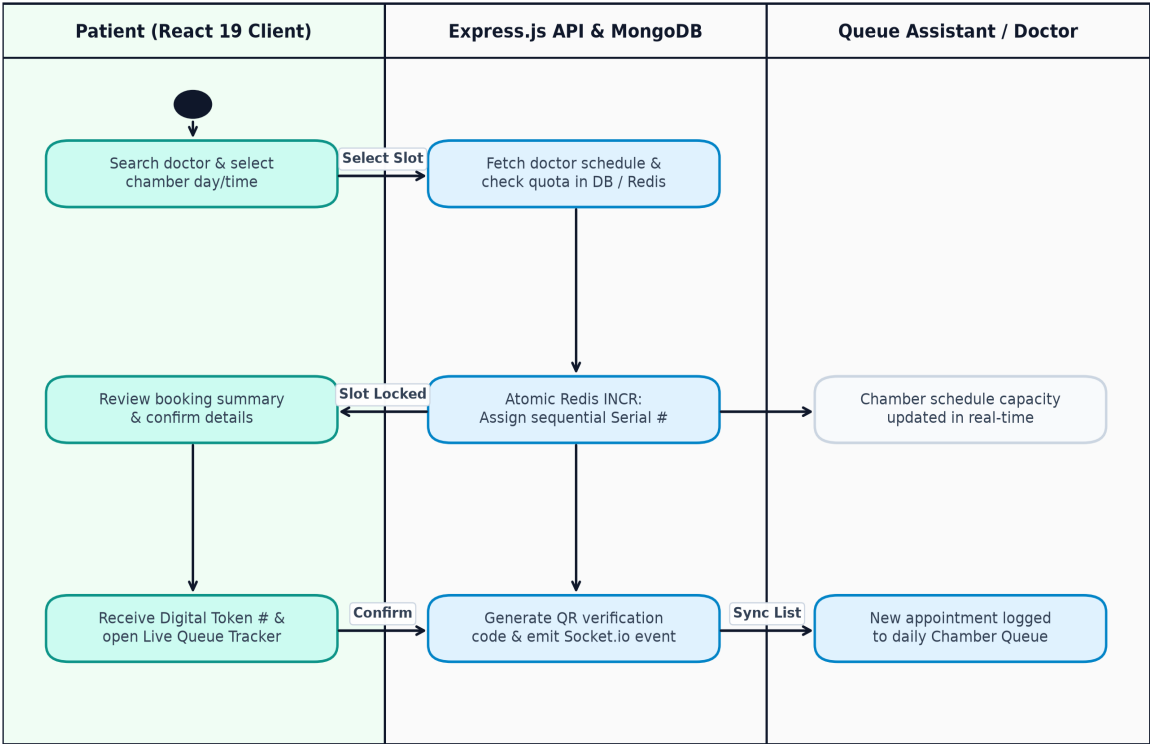


Figure 3.5: Swimlane Diagram for Doctor Appointment & Serial Booking Flow

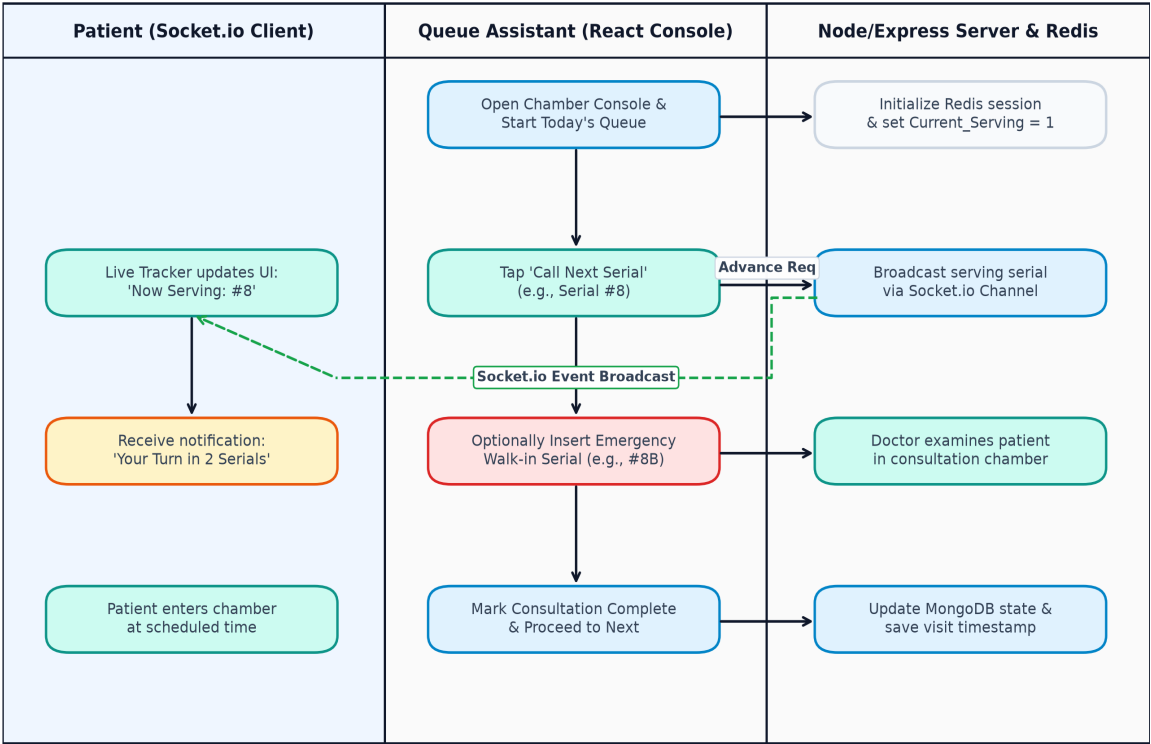


Figure 3.6: Swimlane Diagram for Live Chamber Queue Advancement & Consultation

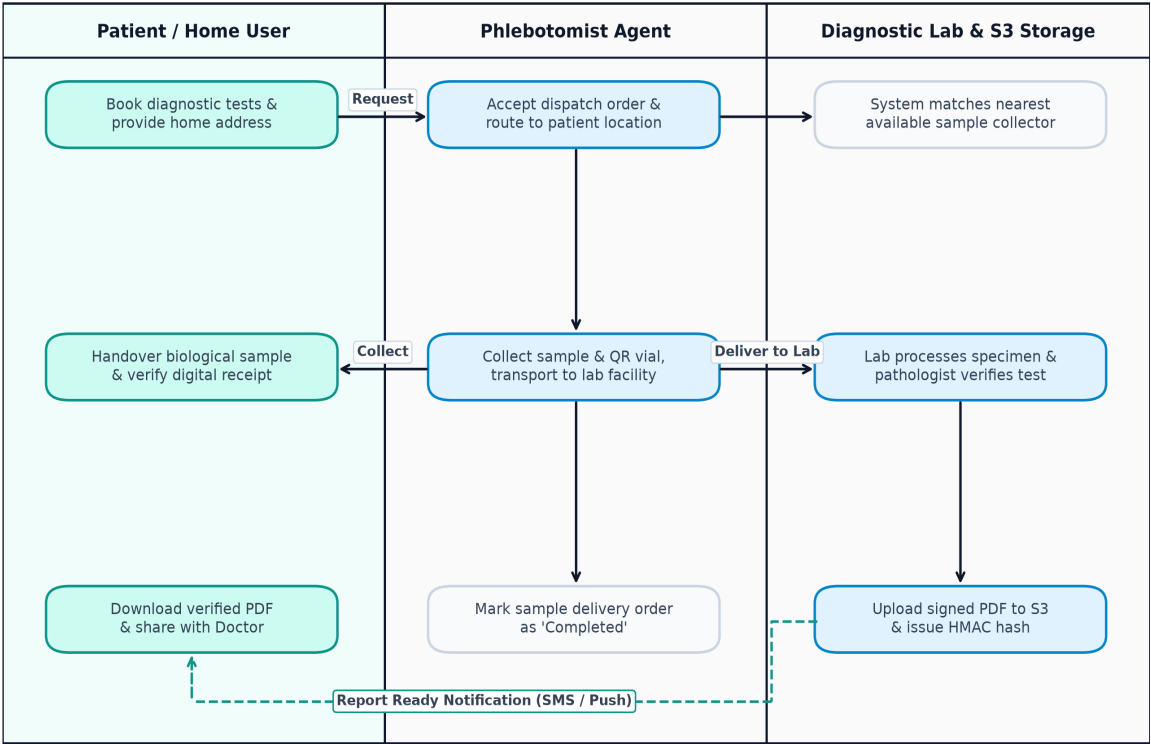


Figure 3.7: Swimlane Diagram for Home Diagnostic Collection & Report Verification

3.7 Class Diagram & Mongoose Schema Architecture

Figure 3.8 details the domain class model and Mongoose document schema design, showing attributes, methods, and relationships across `User`, `Patient`, `Doctor`, `ChamberSchedule`, `Appointment`, `QueueSession`, `DiagnosticOrder`, and `AuditLog`.

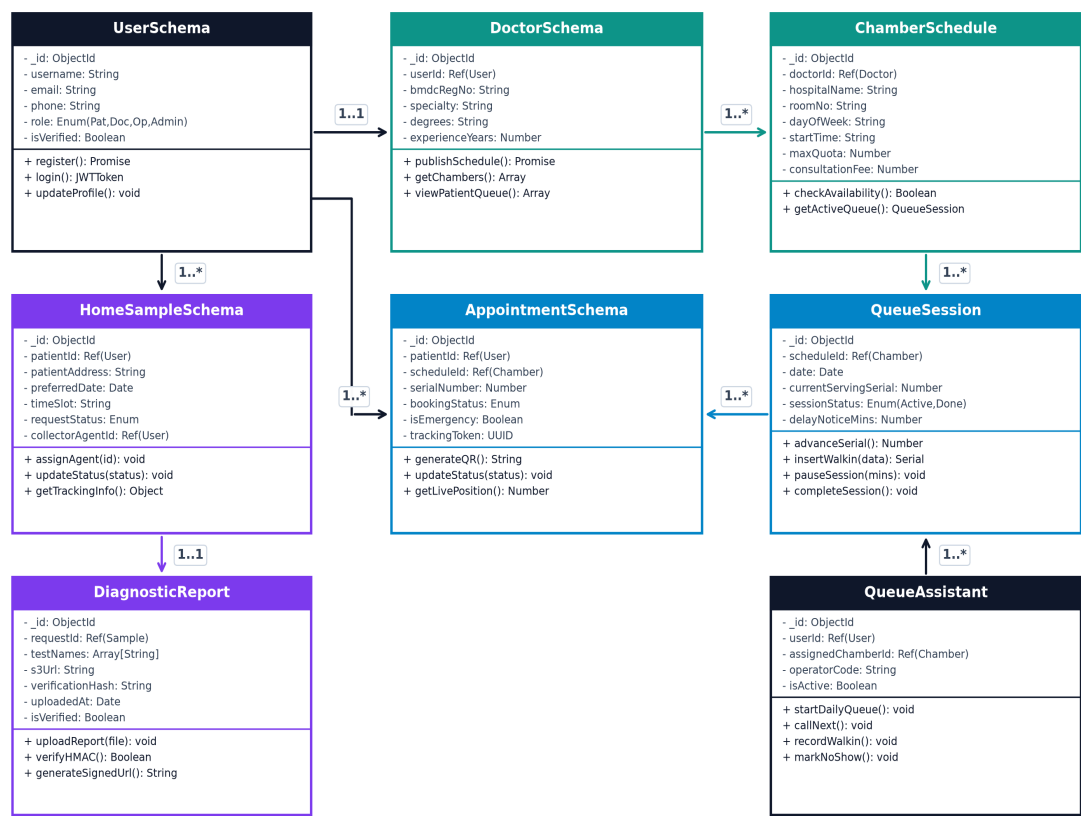


Figure 3.8: Comprehensive UML Class Diagram & Mongoose Schema Architecture

3.8 Sequence Diagrams

Sequence diagrams illustrate time-ordered message passing between client components, REST APIs, Redis in-memory caches, MongoDB, and WebSocket gateways.

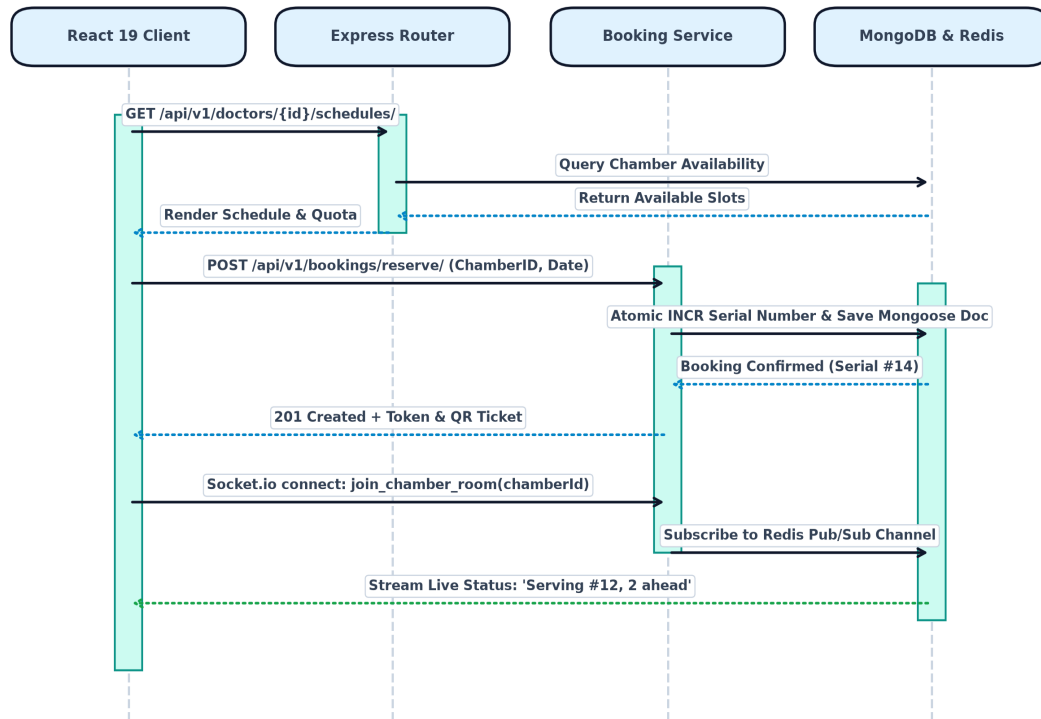


Figure 3.9: Sequence Diagram for Serial Booking & Socket.io Live Queue Stream

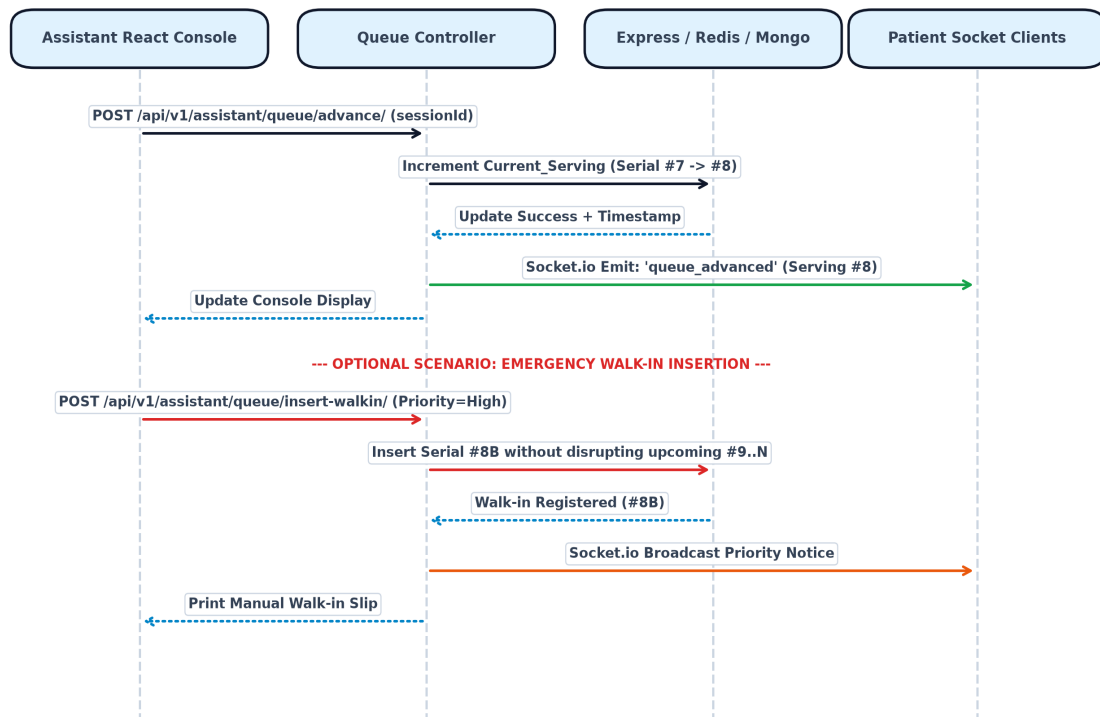


Figure 3.10: Sequence Diagram for Queue Assistant Advancement & Emergency Walk-in

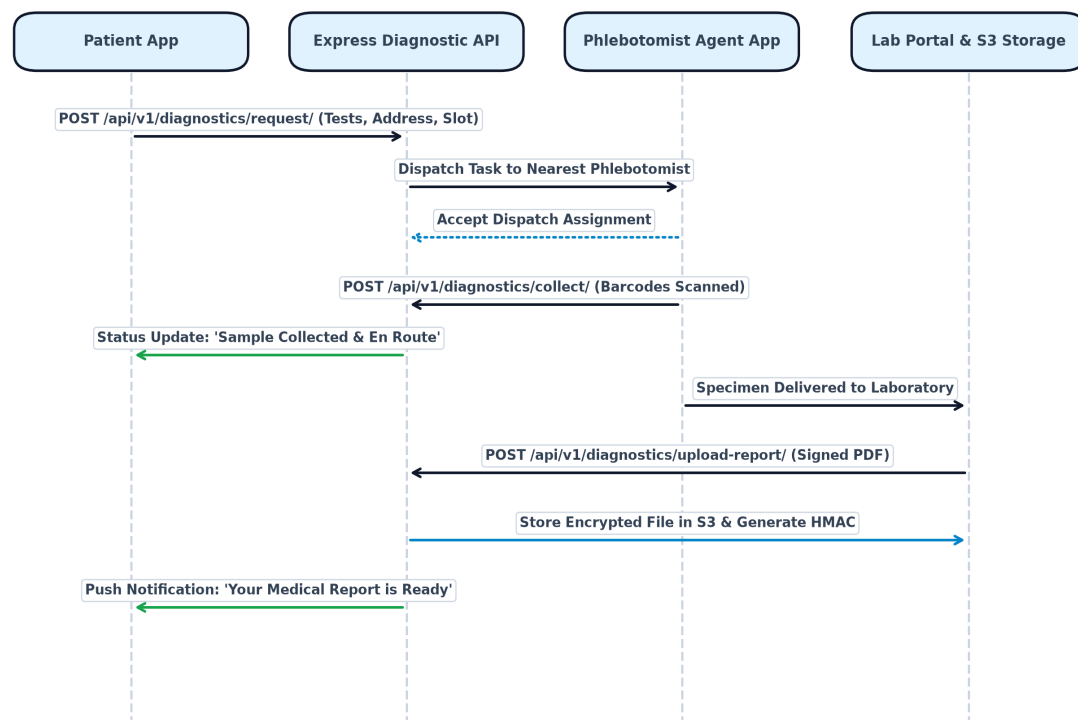


Figure 3.11: Sequence Diagram for Home Sample Request, Collection & Report Delivery

3.9 State Transition Diagrams

State transition diagrams capture lifecycle state mutations and trigger events for core entities in SerialSync.

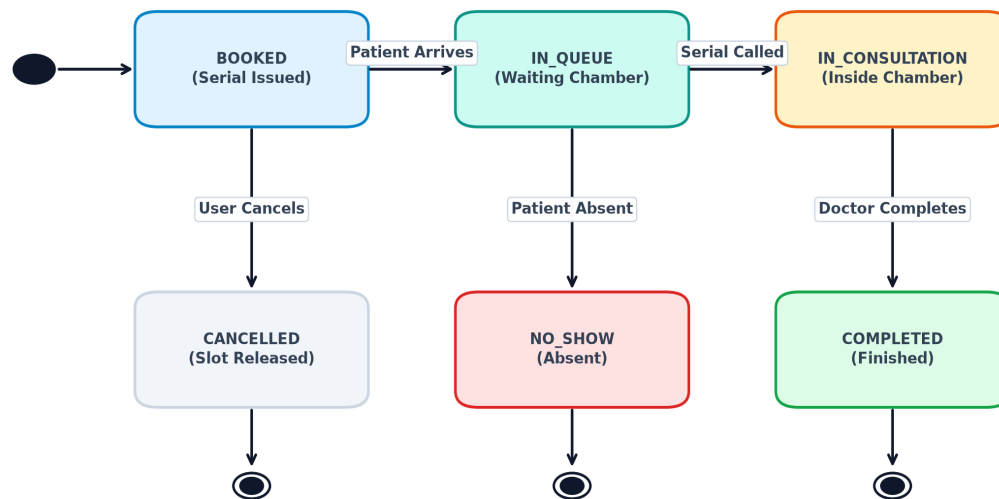


Figure 3.12: State Diagram for Appointment & Serial Booking Lifecycle

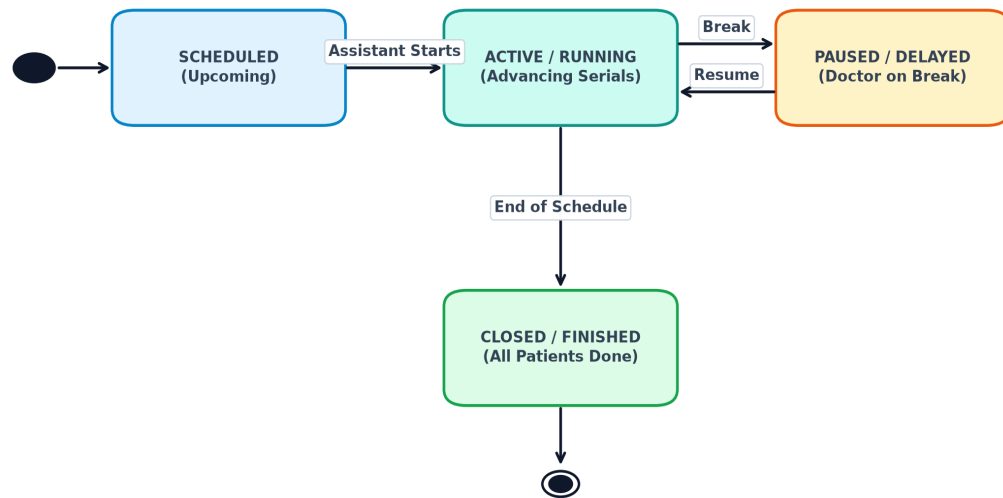


Figure 3.13: State Diagram for Chamber Queue Session Management

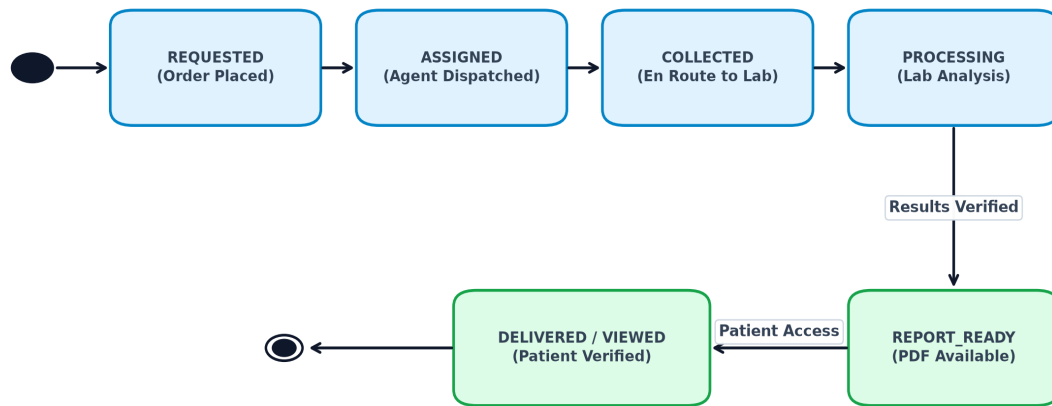


Figure 3.14: State Diagram for Home Diagnostic Sample Collection & Report Lifecycle

3.10 Safety and Security Requirements

3.10.1 Access Requirements (RBAC)

SerialSync enforces strict Role-Based Access Control (RBAC) across five roles: Patient, Doctor, Queue Assistant, Phlebotomist, and Admin. JWT access tokens (15-minute expiry) are stored exclusively in HTTP-only, SameSite=Strict cookies, protecting against Cross-Site Scripting (XSS). All mutation routes are guarded by Express middleware checking role authorization and account verification status.

3.10.2 Integrity Requirements (Atomic Queue Concurrency)

To prevent double-booking during high-concurrency peak morning checkouts, serial allocation relies on atomic Redis `INCRBY`` operations guaranteed by Redis single-threaded in-memory execution. Furthermore, MongoDB document writes are enclosed in multi-document transactions with write concern `majority``, ensuring 100% serialized booking integrity.

3.10.3 Privacy Requirements (Medical Records Encryption)

All uploaded diagnostic laboratory PDF reports are stored in private, non-public AWS S3 buckets encrypted at rest via AES-256. Direct bucket access is blocked; patients and consulting physicians access reports strictly via temporary, 15-minute HMAC-SHA256 presigned URLs generated by the backend after authenticating the user's role and permission grant.

4. Architecture

4.1 Architectural Model/Style Used

SerialSync is architected as an **Event-Driven, Multi-Tier MERN Web Application** combining a stateless RESTful API tier, a real-time WebSocket event pipeline (Socket.io), and an in-memory caching and message brokerage tier (Redis) over a document-oriented database persistence layer (MongoDB Atlas). The architectural style adheres strictly to the **3-Tier Layered Architecture** with asynchronous pub/sub event decoupling for real-time queue synchronization.

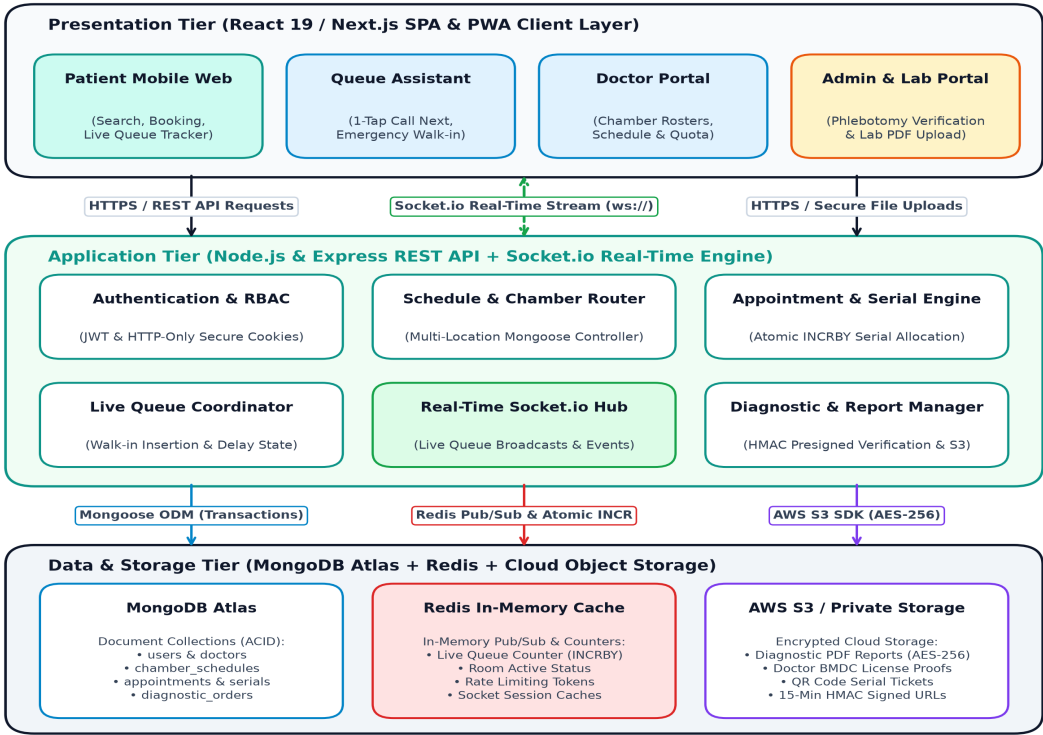


Figure 4.1: SerialSync MERN Stack Multi-Tier Architecture & Real-Time Sync Pipeline

4.1.1 Rationale for Choosing Architectural Model/Style

The selection of the event-driven 3-Tier MERN architecture is grounded in the critical real-time performance requirements of outpatient queuing in high-density urban clinics. Traditional request-response polling (e.g. AJAX polling every 5 seconds) imposes unsustainable server overhead with $O(N)$ network traffic scaling per active waiting patient. Socket.io WebSockets establish persistent full-duplex TCP channels, reducing queue state broadcast latency from >5000ms to <150ms with minimal bandwidth overhead.

4.2 Technology, Software, and Hardware Used

Architectural Layer	Core Technology	Key Responsibilities & Architectural Role
Presentation Tier	React 19 / Next.js SPA, TailwindCSS, Socket.io Client	High-performance responsive web and mobile PWA clients for Patients, Doctors, Queue Assistants, and Administrators with optimistic UI updates.
Application Tier	Node.js (v20 LTS), Express.js Micro-Routers	Stateless REST API microservices, JWT authentication, RBAC authorization middleware, atomic sequential serial allocation engine, and diagnostic report generator.
Real-Time Event Hub	Socket.io WebSocket Server, Redis Pub/Sub	Bi-directional real-time event broadcasting for live chamber queue state ('Now Serving: #12'), delay notifications, and emergency walk-in broadcasts.
Data & Storage Tier	MongoDB Atlas, Mongoose ODM, AWS S3	Document-oriented ACID multi-document transactional persistence for users, schedules, appointments, and HMAC-SHA256 encrypted digital diagnostic reports.

Table 4.1: MERN Stack Component Responsibilities Breakdown for SerialSync

5. Design

5.1 Component Level Design Following Pattern

SerialSync enforces strict software design patterns across its core subsystems: **Strategy Pattern** for multi-role RBAC authorization, **Factory Pattern** for dynamic chamber slot generation, **Observer & Pub/Sub Pattern** for real-time WebSocket queue synchronization, and **Template Method Pattern** for the multi-stage diagnostic specimen lifecycle.

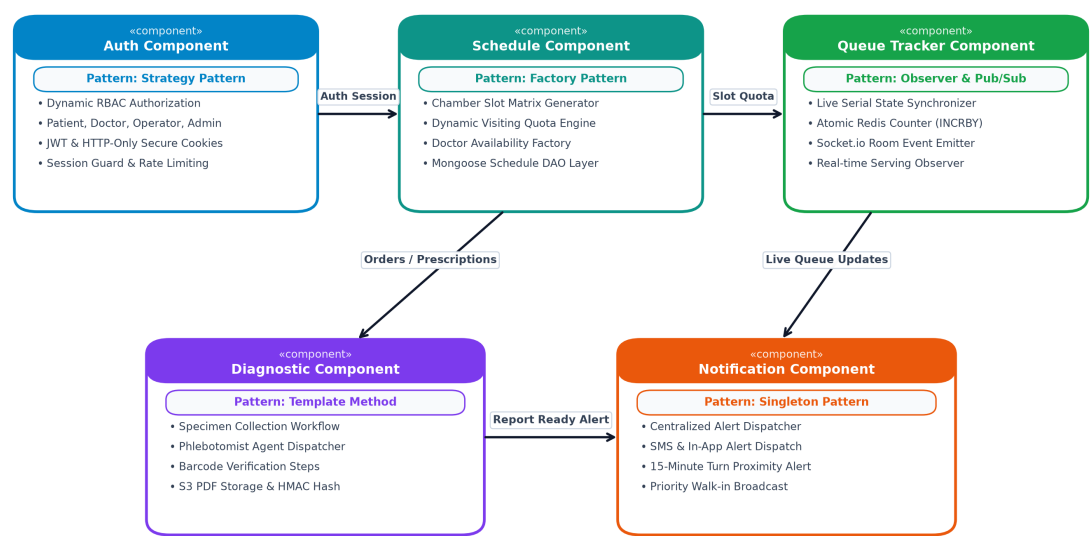


Figure 5.1: Component-Level Design Architecture Following Design Patterns (MERN)

5.2 Dataflow Diagram

Data Flow Diagrams depict how data enters, is processed, and is stored within SerialSync across three levels of decomposition.

5.2.1 DFD Level 0 (System Context Diagram)

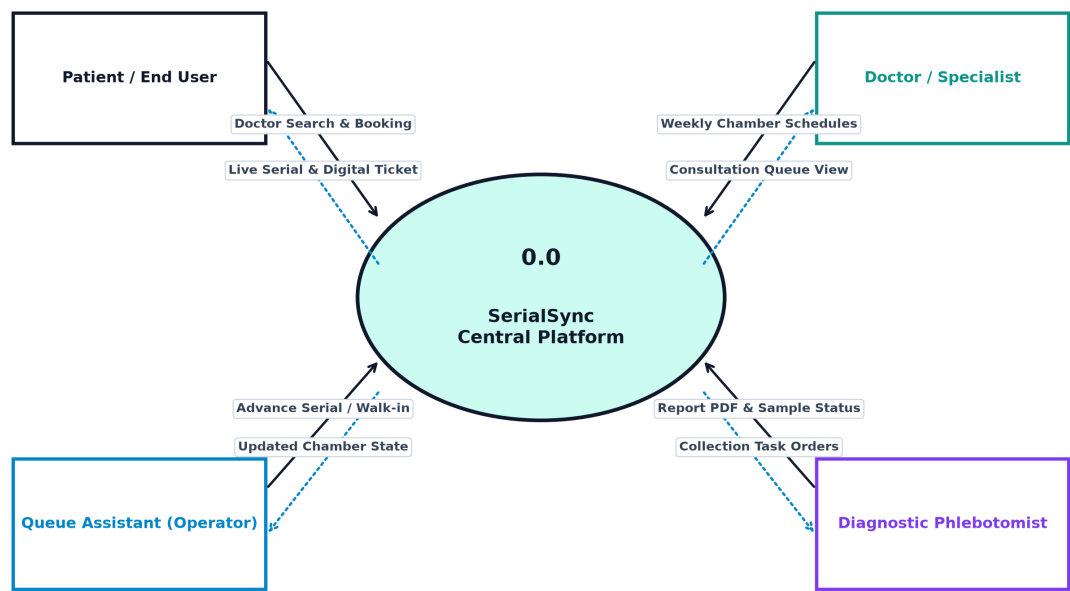


Figure 5.2: Data Flow Diagram (DFD) Level 0 - System Context Diagram

5.2.2 DFD Level 1 (Functional Decomposition)

Figure 5.3 shows the functional decomposition into 5 core processes: 1.0 Auth & Profile, 2.0 Doctor Chamber Scheduling, 3.0 Serial Booking Engine, 4.0 Live Queue Synchronization, and 5.0 Diagnostic Specimen & Report Handling.

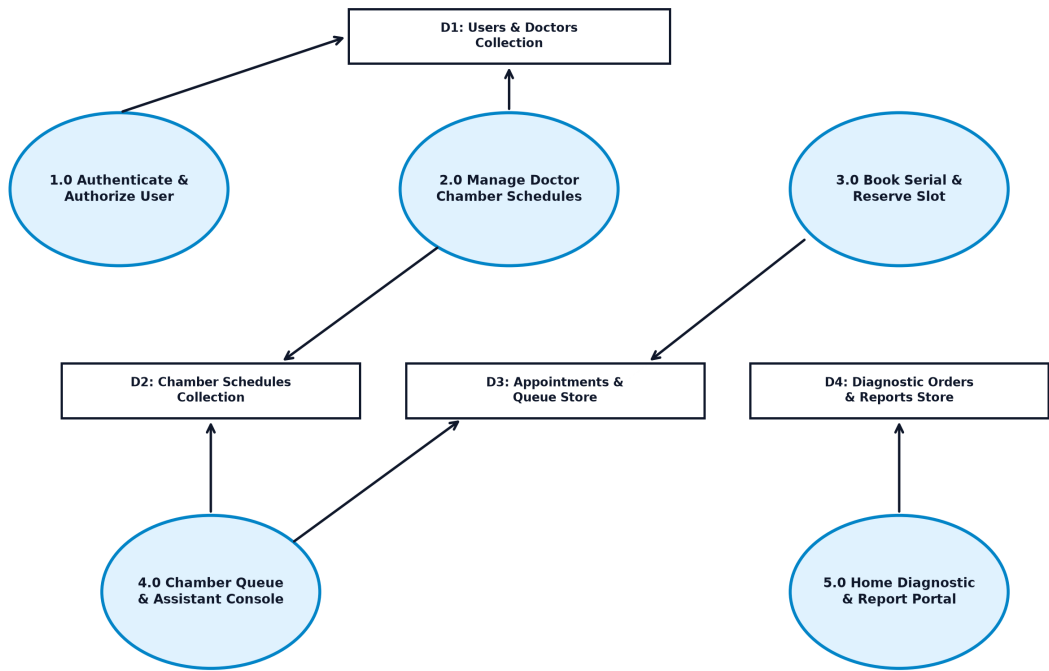


Figure 5.3: Data Flow Diagram (DFD) Level 1 - System Functional Decomposition

5.2.3 DFD Level 2 (Real-Time Queue & Chamber Console Decomposition)

Figure 5.4 details sub-processes for Process 4.0: 4.1 (Validate Assistant Auth), 4.2 (Advance Serving Counter in Redis), 4.3 (Insert Emergency Walk-in Token), and 4.4 (Broadcast Socket.io Event to Patient Room).

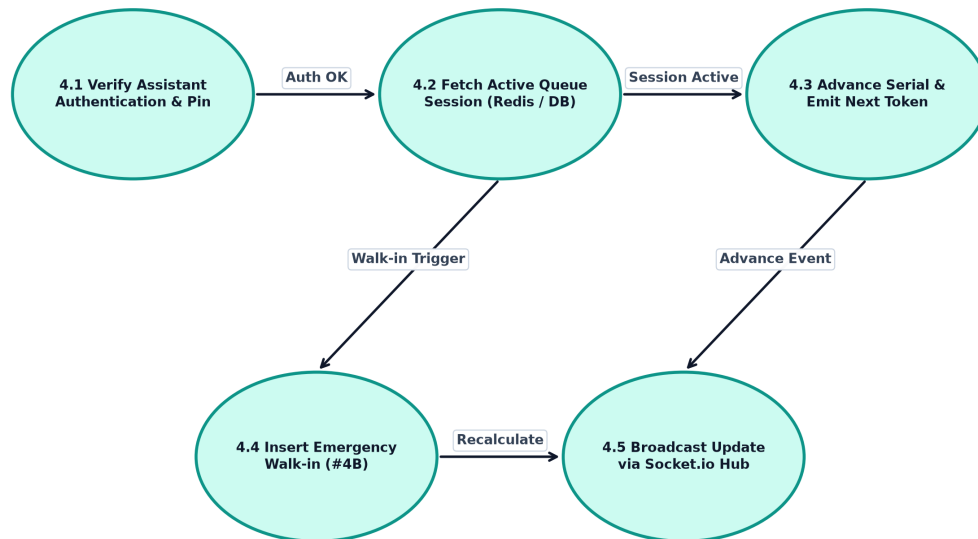


Figure 5.4: Data Flow Diagram (DFD) Level 2 - Live Queue & Walk-in Synchronization

5.2.1 Entity Relationship Diagram

Figure 5.5 presents the comprehensive MongoDB document collection schema and entity relationship model, illustrating relations between `User`, `Doctor`, `ChamberSchedule`, `Appointment`, `QueueSession`, `DiagnosticOrder`, and `AuditLog`.

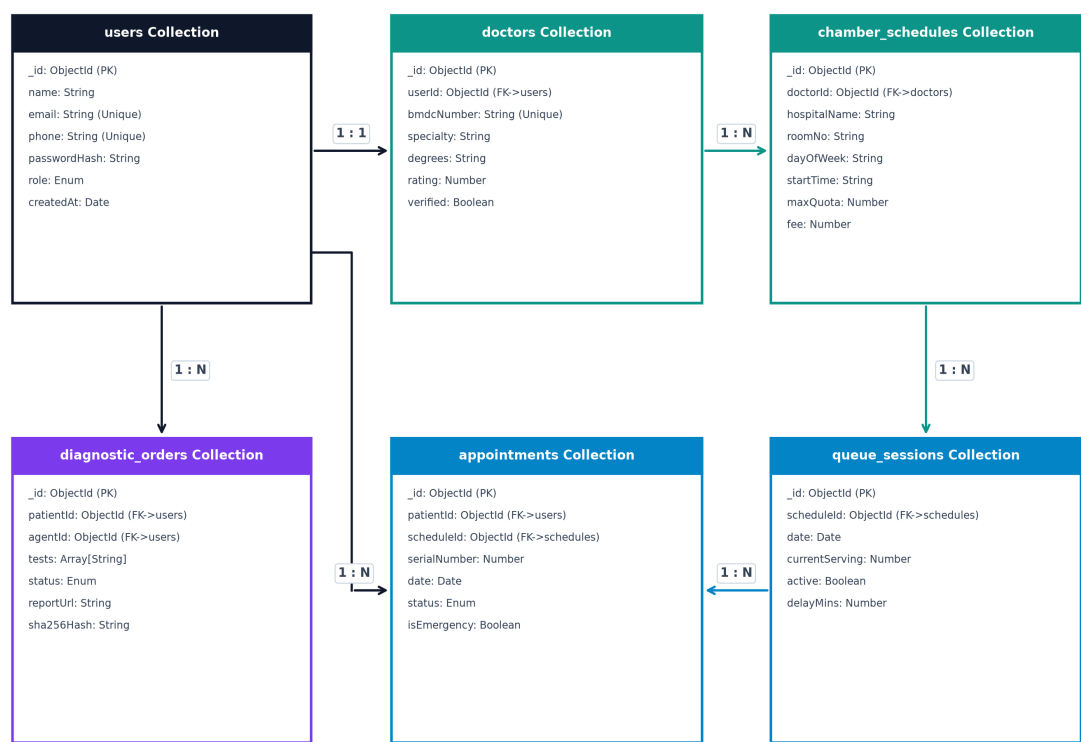


Figure 5.5: MongoDB Document Collection Schema Architecture (MERN)

5.3 GUI (Graphical User Interface) design

The user interface of SerialSync is engineered following a modern, high-contrast dark medical aesthetic (Slate-900 / Sky-600 / Emerald-500) inspired by leading on-demand platforms (Pathao, Uber Health, Linear, Practo). The design incorporates responsive grid layouts, glassmorphism cards, live data badges, real-time WebSocket state synchronizers, and touch-optimized ergonomics for high-velocity clinic operations.

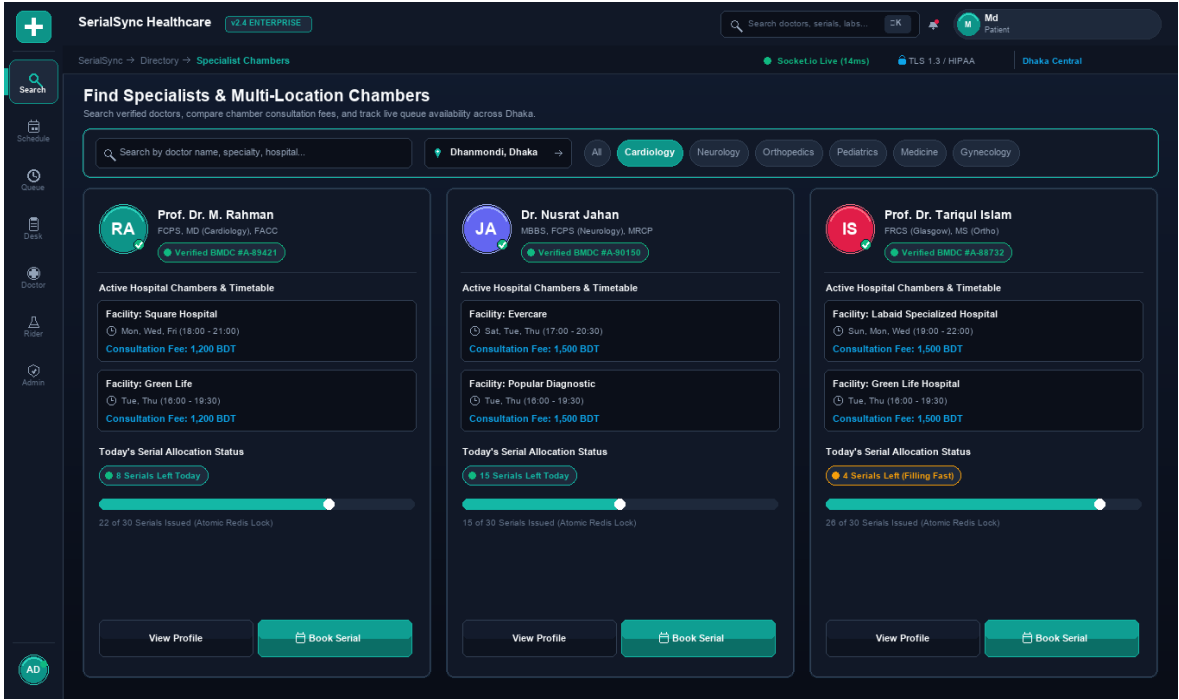


Figure 5.6: GUI Mockup - Patient Specialist Search & Multi-Location Chamber Portal (React 19 / Dark Theme)

Patient Specialist Search & Multi-Location Chamber Portal: Features dynamic auto-completion for medical specialties (Cardiology, Neurology, Orthopedics, Pediatrics), Dhaka city area filters (Dhanmondi, Gulshan, Uttara), doctor profile cards with BMDC verified badges (#A-89421), multi-hospital affiliations (Square Hospital, Green Life, LabAid), consultation fees, live active quota counters ('8 Serials Left Today'), and instant 'Book Serial / View Schedule' CTAs.

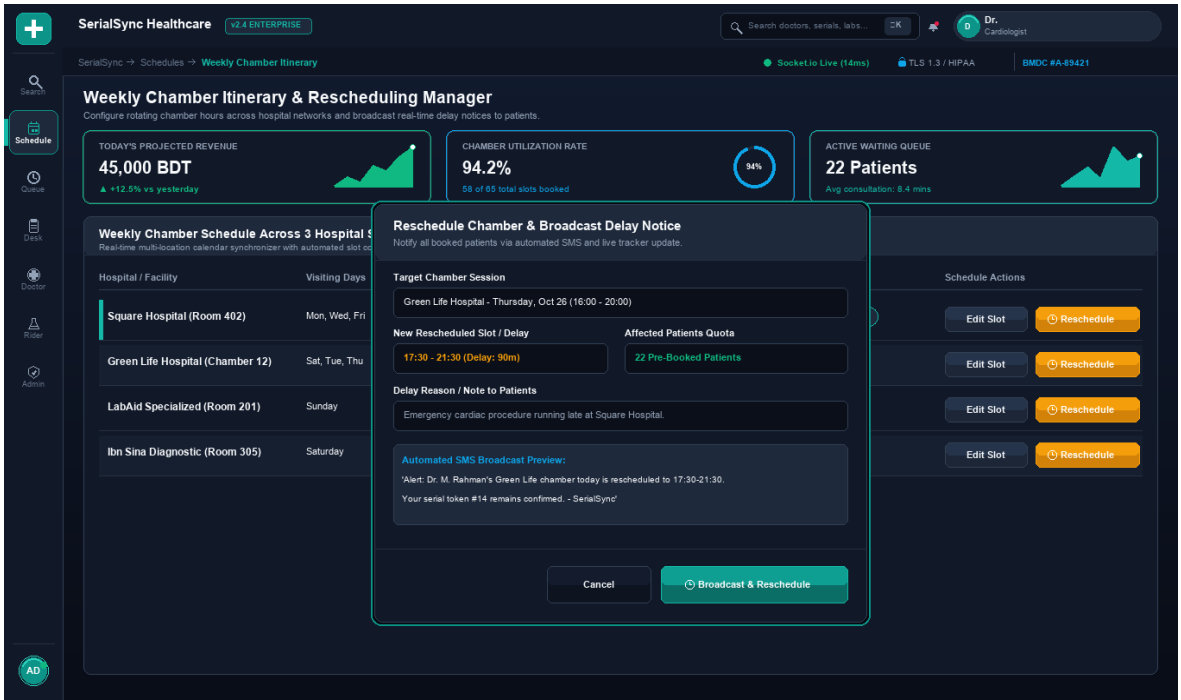


Figure 5.7: GUI Mockup - Doctor Multi-Location Schedule & Rescheduling Portal

Doctor Multi-Location Schedule & Rescheduling Portal: Displays the specialist's weekly schedule across multiple hospitals (Square, Green Life, LabAid) alongside real-time occupancy metrics (94%) and earnings. Includes an interactive 'Reschedule Chamber / Delay Notification' modal with date picker, reason selector, and automated SMS/Email broadcast dispatch to all 22 booked patients.

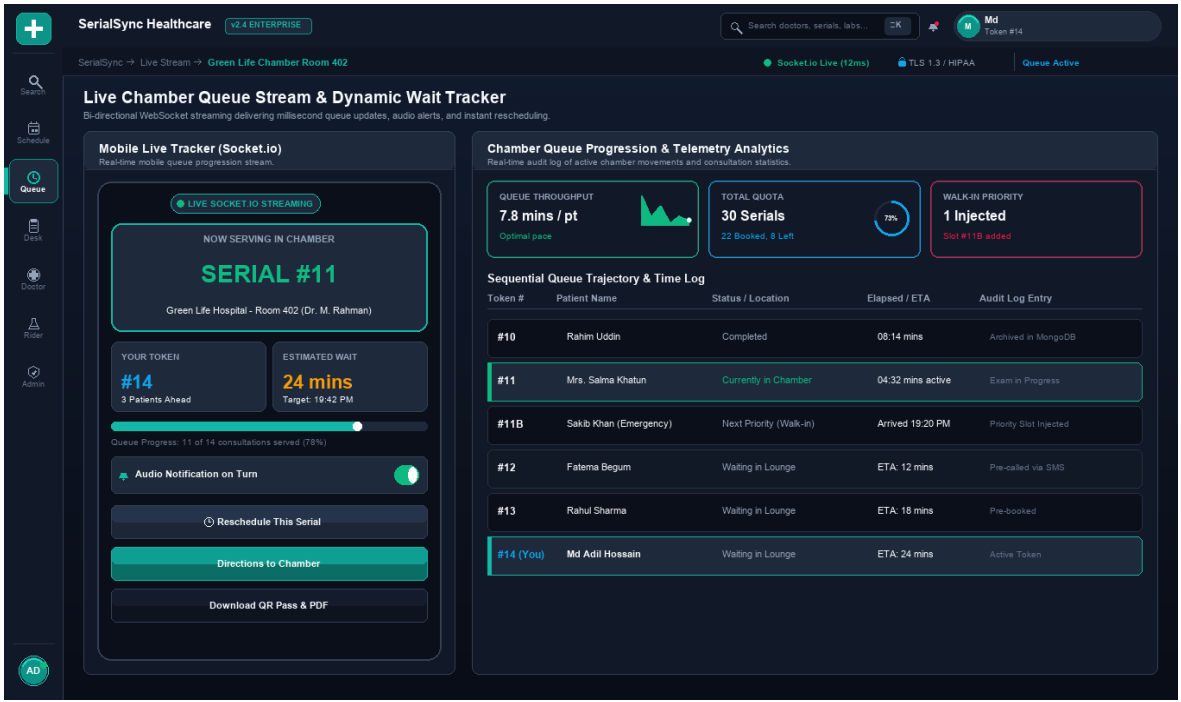


Figure 5.8: GUI Mockup - Patient Real-Time Live Queue Tracker (Socket.io Powered)

Patient Real-Time Live Queue Tracker: Responsive mobile tracking interface featuring live Socket.io WebSocket streaming badge, glowing 'Now Serving: Serial #11 (Room 402, Green Life)', booked patient token '#14 (3 Patients Ahead)', dynamic estimated wait countdown (24 mins), audio chime toggle, and 1-tap 'Reschedule Serial' and 'Directions to Chamber' action buttons.

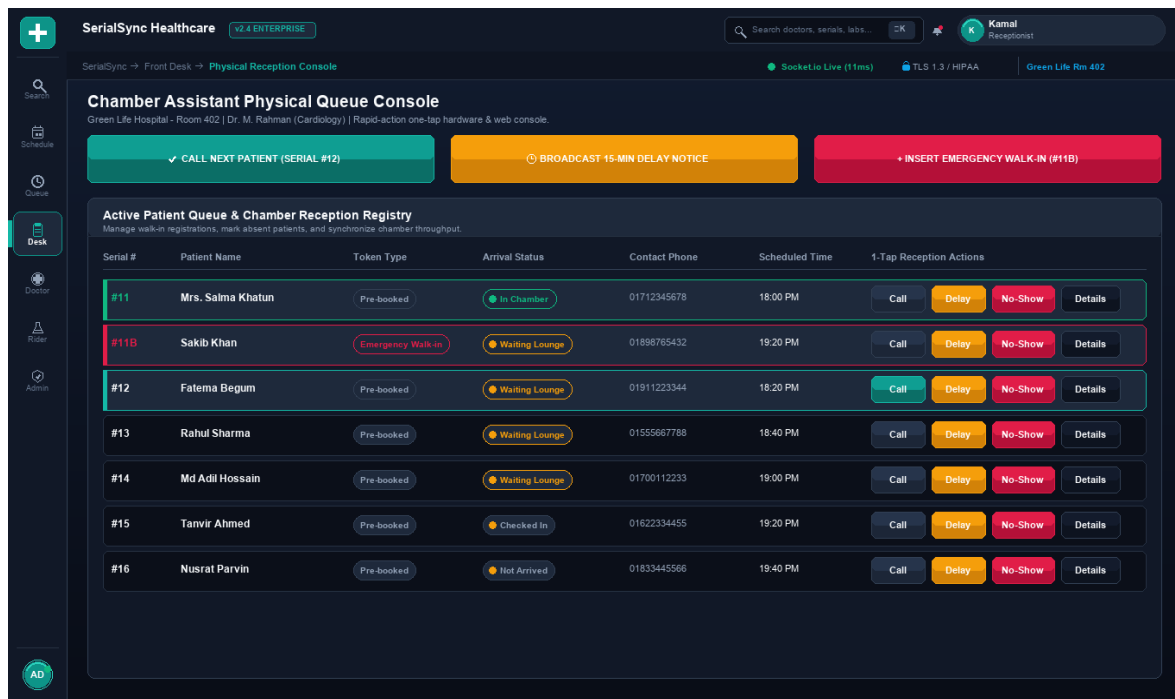


Figure 5.9: GUI Mockup - Queue Assistant Physical Chamber Control Console

Queue Assistant Physical Chamber Control Console: High-contrast, rapid-action control panel used by physical clinic receptionists. Features prominent 1-tap 'Call Next (Serial #12)' (green), 'Broadcast 15-min Delay Alert' (amber), and '+ Insert Emergency Walk-in Serial (#11B)' (red) buttons, alongside an active patient queue table with instant Call, Delay, Skip, and View Details actions.

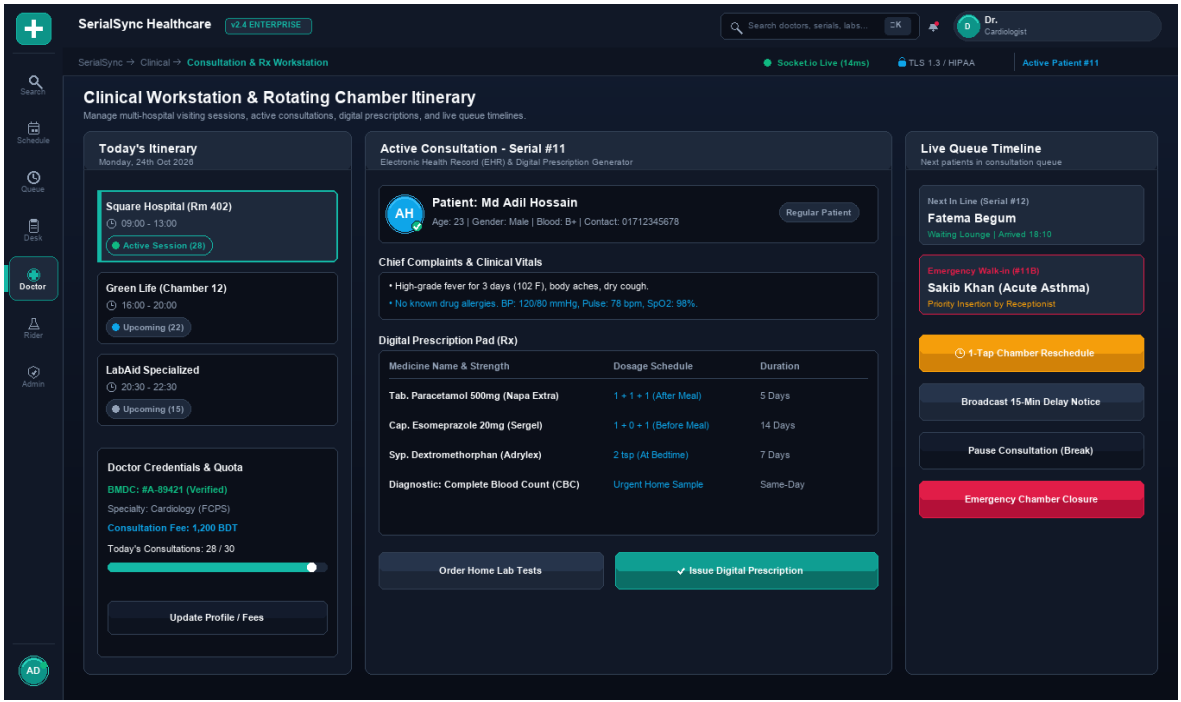


Figure 5.10: GUI Mockup - Doctor Consultation & Multi-Hospital Rotating Itinerary

Doctor Consultation & Multi-Hospital Rotating Itinerary: Dedicated clinical dashboard displaying today's rotating multi-hospital itinerary (Square Hospital 09:00-13:00 ACTIVE, Green Life 16:00-20:00, LabAid 20:30-22:30), active patient consultation records (Serial #11 - Md Adil Hossain with chief complaint, medical history, and digital prescription pad), upcoming queue timeline, instant delay notification trigger, and 1-tap reschedule tool.

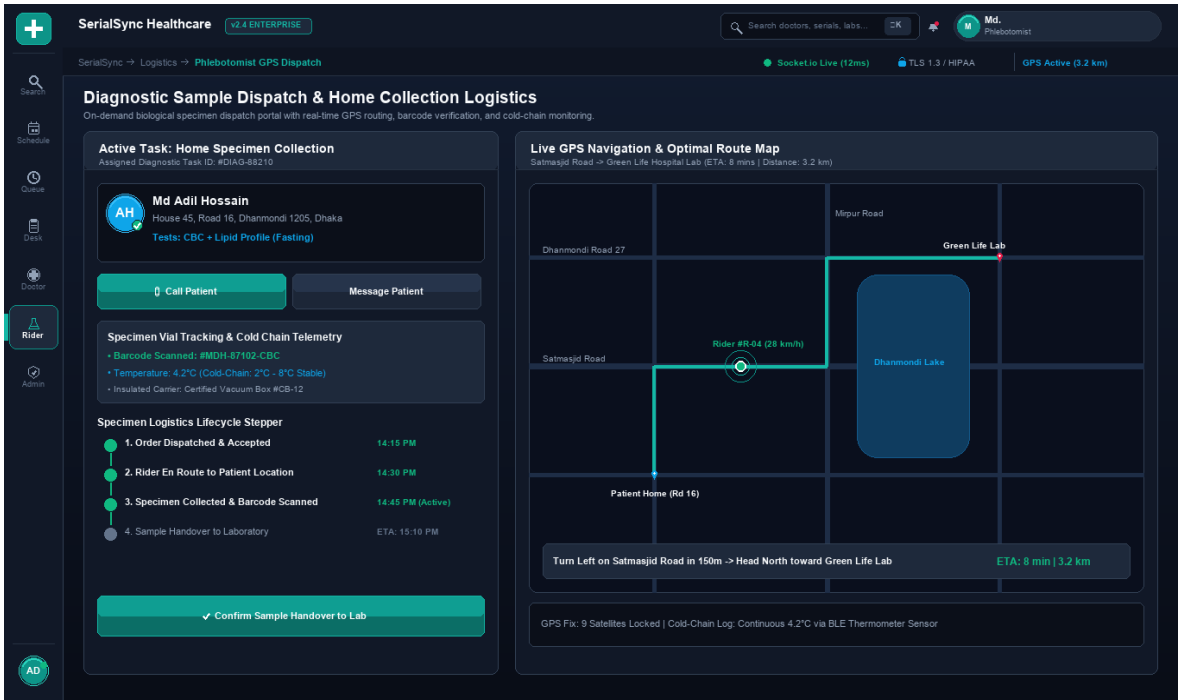


Figure 5.11: GUI Mockup - Pathao-Style Phlebotomist Rider & Sample Dispatch Portal

Pathao-Style Phlebotomist Rider & Sample Dispatch Portal: Logistics and home specimen collection mobile interface for field diagnostic agents. Features active pickup tasks (Patient: Md Adil Hossain, Dhanmondi 1205, Dhaka), live GPS map routing (Satmasjid Rd / Green Life Hospital, ETA: 8 min), 1-tap 'Call Patient' and 'Message' communication buttons, biological specimen vial barcode scanner (#MDH871), cold-chain temperature monitor (4.2°C stable), and a real-time 4-stage collection stepper (Assigned -> En Route -> Sample Collected -> Delivered to Lab).

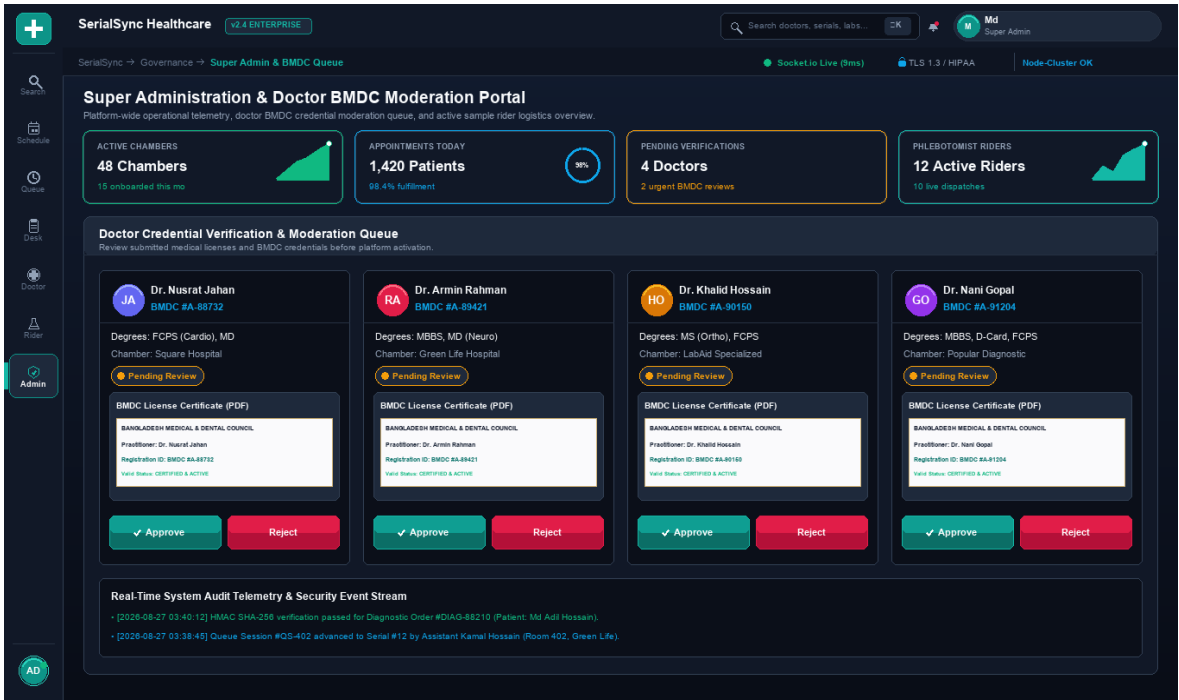


Figure 5.12: GUI Mockup - Super Administration & Doctor BMDC Moderation Portal

Super Administration & Doctor BMDC Moderation Portal: Platform governance console featuring real-time operational telemetry (48 active chambers, 1,420 appointments, 12 active riders), interactive Doctor Credential Moderation Queue with medical license PDF preview and 1-tap 'Approve & Activate' / 'Reject with Reason' actions, and a live city-wide GPS fleet map tracking active sample collection riders.

6. API Documentation

SerialSync exposes a modern, high-throughput RESTful API along with a bi-directional WebSocket interface (Socket.io) operating over HTTPS and WSS protocols. All endpoints consume and produce JSON (`application/json`). Standard HTTP status codes are strictly enforced across all microservices: `200 OK`, `201 Created`, `400 Bad Request`, `401 Unauthorized`, `403 Forbidden`, `404 Not Found`, `409 Conflict`, and `500 Internal Server Error`.

6.1 Module 1: Authentication & User Management

POST /api/v1/auth/register/ Access: Public (All Users)

Description: Registers a new user account (Patient, Doctor, or Queue Assistant) with phone number deduplication and password hashing using bcrypt (12 salt rounds).

Parameter	Type	Required	Description
username	String	Yes	Full legal name of the user.
email	String	Yes	Unique email address for notifications.
phone	String	Yes	Unique 11-digit Bangladeshi mobile number (01XXXXXXXX).
password	String	Yes	Minimum 8-character password containing letters and numbers.
role	Enum	Yes	User role: `patient`, `doctor`, `assistant`, `agent`.
Sample Request Body		Sample JSON Response (200 / 201)	
<pre>{ "username": "Md Adil Hossain", "email": "adil@example.com", "phone": "01712345678", "password": "SecurePassword123", "role": "patient" }</pre>		<pre>{ "status": "success", "statusCode": 201, "data": { "userId": "usr_99210", "username": "Md Adil Hossain", "role": "patient", "token": "eyJhbGciOiJIUzI1Ni..." } }</pre>	

POST /api/v1/auth/login/ Access: Public (All Users)

Description: Authenticates existing users via phone and password; returns a cryptographically signed JWT access token in an HTTP-only cookie.

Parameter	Type	Required	Description
phone	String	Yes	Registered 11-digit mobile phone number.
password	String	Yes	Account plaintext password.
Sample Request Body		Sample JSON Response (200 / 201)	
<pre>{ "phone": "01712345678", "password": "SecurePassword123" }</pre>		<pre>{ "status": "success", "statusCode": 200, "data": { "userId": "usr_99210", "role": "patient", "expiresIn": 86400 } }</pre>	

6.1 Module 1: Authentication & User Management (Continued)

GET /api/v1/users/me/			Access: Authenticated User
Description: Retrieves the currently logged-in user profile, permissions, active booking count, and verified credentials.			
Parameter	Type	Required	Description
Authorization	Header	Yes	Bearer
fields	Query	No	Optional projection fields (`profile`, `stats`).
Sample Request Body		Sample JSON Response (200 / 201)	
None (GET Request with Cookie/Header)		<pre>{ "status": "success", "data": { "id": "usr_99210", "name": "Md Adil Hossain", "phone": "01712345678", "role": "patient", "verified": true, "totalBookings": 4 } }</pre>	

6.2 Module 2: Doctor Multi-Location Schedules

POST /api/v1/doctors/schedule/create/

Access: Verified Doctor

Description: Creates a new recurring weekly chamber schedule at a specific hospital or private clinic with automated conflict detection.

Parameter	Type	Required	Description
hospitalName	String	Yes	Full clinic or hospital facility name.
roomNo	String	Yes	Chamber room number or floor location.
dayOfWeek	Enum	Yes	Day: `Mon`, `Tue`, `Wed`, `Thu`, `Fri`, `Sat`, `Sun`.
startTime	String	Yes	Consultation start time in 24h format ('18:00').
endTime	String	Yes	Consultation end time in 24h format ('21:00').
maxQuota	Integer	Yes	Maximum daily patient capacity (1–100).
fee	Number	Yes	Consultation fee in BDT.
Sample Request Body		Sample JSON Response (200 / 201)	
<pre>{ "hospitalName": "Green Life Hospital", "roomNo": "Room 402", "dayOfWeek": "Mon", "startTime": "18:00", "endTime": "21:00", "maxQuota": 30, "fee": 1200 }</pre>		<pre>{ "status": "success", "statusCode": 201, "data": { "scheduleId": "sch_402", "status": "ACTIVE" } }</pre>	

GET /api/v1/doctors/search/

Access: Public (All Users)

Description: Filters and discovers doctors across hospitals based on specialty, degrees, city area, and available day.

Parameter	Type	Required	Description
specialty	String	No	Medical specialty (e.g., 'Cardiology').
area	String	No	Geographic area (e.g., 'Dhanmondi').
day	String	No	Target day of week ('Mon').
Sample Request Body		Sample JSON Response (200 / 201)	
None (Query Parameters)		<pre>{ "status": "success", "results": 1, "data": [{ "doctorId": "doc_101", "name": "Prof. Dr. M. Rahman", "specialty": "Cardiology", "chambers": [{ "hospital": "Green Life", "fee": 1200 }] }] }</pre>	

6.2 Module 2: Doctor Multi-Location Schedules (Continued)

PATCH /api/v1/doctors/schedule/:id/			Access: Verified Doctor
Description: Modifies chamber parameters (fee, quota, room) or temporarily deactivates a specific hospital schedule.			
Parameter	Type	Required	Description
scheduleId	Path	Yes	Unique MongoDB schedule ObjectId.
fee	Number	No	Updated consultation fee.
active	Boolean	No	Schedule visibility toggle.
Sample Request Body		Sample JSON Response (200 / 201)	
{ "fee": 1500, "active": true }		{ "status": "success", "message": "Schedule updated successfully" }	

6.3 Module 3: Appointment Booking & Dynamic Serial Allocation

POST /api/v1/appointments/reserve/

Access: Authenticated Patient

Description: Atomically reserves a sequential serial token using Redis `INCR` locks and creates an Appointment record.

Parameter	Type	Required	Description
scheduleId	String	Yes	Target ChamberSchedule ObjectID.
appointmentDate	Date	Yes	ISO Date string ('2026-09-14').
patientName	String	Yes	Consulting patient legal name.
patientAge	Integer	Yes	Patient age in years.
Sample Request Body		Sample JSON Response (200 / 201)	
<pre>{ "scheduleId": "sch_402", "appointmentDate": "2026-09-14", "patientName": "Md Adil Hossain", "patientAge": 23 }</pre>		<pre>{ "status": "success", "statusCode": 201, "data": { "appointmentId": "apt_99812", "serialNo": 14, "qrPassUrl": "https://cdn.serialsync.com/qr/apt_99812.png" } }</pre>	

GET /api/v1/appointments/my-bookings/

Access: Authenticated Patient

Description: Returns a chronological list of past and upcoming consultation bookings for the authenticated patient.

Parameter	Type	Required	Description
status	String	No	Filter by `CONFIRMED`, `COMPLETED`, `CANCELLED`.
Sample Request Body		Sample JSON Response (200 / 201)	
None (GET Request)		<pre>{ "status": "success", "data": [{ "appointmentId": "apt_99812", "doctor": "Prof. Dr. M. Rahman", "serialNo": 14, "status": "CONFIRMED" }] }</pre>	

6.3 Module 3: Appointment Booking & Dynamic Serial Allocation (Continued)

DELETE /api/v1/appointments/:id/cancel/			Access: Authenticated Patient
Description: Cancels an upcoming serial booking (>2h before chamber start), releasing quota back to the pool via Redis `DECR`.			
Parameter	Type	Required	Description
appointmentId	Path	Yes	Appointment ObjectId to cancel.
Sample Request Body		Sample JSON Response (200 / 201)	
None (DELETE Request)		{ "status": "success", "message": "Appointment cancelled. Quota released." }	

6.4 Module 4: Queue Assistant & Chamber Operations

POST /api/v1/assistant/queue/advance/

Access: Chamber Assistant

Description: Advances the active serving serial counter by +1; emits live `queue\_updated` WebSocket event to patient room.

Parameter	Type	Required	Description
sessionId	String	Yes	Active QueueSession ObjectId.
notes	String	No	Optional consultation notes.
Sample Request Body		Sample JSON Response (200 / 201)	
<pre>{ "sessionId": "qses_301", "notes": "Patient 11 consultation finished" }</pre>		<pre>{ "status": "success", "data": { "currentServingSerial": 12, "patientsRemaining": 18, "estimatedWaitMins": 144 } }</pre>	

POST /api/v1/assistant/queue/walkin/

Access: Chamber Assistant

Description: Injects an emergency or walk-in patient with priority sub-indexing (e.g. Serial #11B) without disrupting subsequent serials.

Parameter	Type	Required	Description
sessionId	String	Yes	Active QueueSession ObjectId.
patientName	String	Yes	Walk-in patient name.
priority	Enum	Yes	`EMERGENCY`, `URGENT`, `STANDARD`.
Sample Request Body		Sample JSON Response (200 / 201)	
<pre>{ "sessionId": "qses_301", "patientName": "Rahim Uddin", "priority": "EMERGENCY" }</pre>		<pre>{ "status": "success", "data": { "assignedSerial": "11B", "position": "IMMEDIATE_NEXT" } }</pre>	

6.4 Module 4: Queue Assistant & Chamber Operations (Continued)

POST /api/v1/assistant/queue/delay-notice/**Access:** Chamber Assistant**Description:** Broadcasts doctor delay notices (+15m, +30m, +45m) to all waiting patients via WebSocket and SMS.

Parameter	Type	Required	Description
sessionId	String	Yes	Active QueueSession ObjectID.
delayMinutes	Integer	Yes	Estimated delay duration in minutes (15, 30, 45).
reason	String	Yes	Doctor emergency surgery / road traffic.
Sample Request Body		Sample JSON Response (200 / 201)	
{ "sessionId": "qses_301", "delayMinutes": 30, "reason": "Traffic on Mirpur Road" }		{ "status": "success", "message": "Delay broadcast sent to 24 waiting patients" }	

6.5 Module 5: Home Diagnostics & Reports

POST /api/v1/diagnostics/orders/create/

Access: Authenticated Patient

Description: Creates a home diagnostic collection request with selected tests, phlebotomist dispatch, and residential address.

Parameter	Type	Required	Description
testIds	Array	Yes	Array of DiagnosticTest ObjectIds (e.g. CBC, Lipid).
address	String	Yes	Detailed residential collection address.
collectionDate	Date	Yes	Target sample collection date.
timeSlot	String	Yes	Morning time slot ('08:00 - 10:00').
Sample Request Body		Sample JSON Response (200 / 201)	
<pre>{ "testIds": ["test_cbc_01", "test_lipid_02"], "address": "House 14, Road 5, Dhanmondi", "collectionDate": "2026-09-15", "timeSlot": "08:00 - 10:00" }</pre>		<pre>{ "status": "success", "data": { "orderId": "diag_order_771", "status": "PENDING_DISPATCH", "totalAmount": 1600 } }</pre>	

POST /api/v1/diagnostics/reports/upload/

Access: Diagnostic Lab

Description: Uploads finalized PDF lab report to private AWS S3; attaches SHA-256 integrity hash to DiagnosticOrder document.

Parameter	Type	Required	Description
orderId	String	Yes	Target DiagnosticOrder ObjectId.
reportFile	File	Yes	PDF multipart binary file upload.
pathologistId	String	Yes	Signing pathologist license ID.
Sample Request Body		Sample JSON Response (200 / 201)	
Multipart Form-Data (file=report_771.pdf, orderId=diag_order_771)		<pre>{ "status": "success", "data": { "sha256Hash": "e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855", "status": "COMPLETED" } }</pre>	

6.5 Module 5: Home Diagnostics & Reports (Continued)

GET /api/v1/diagnostics/reports/:id/download/			Access: Authorized User
Description: Generates a secure, 15-minute time-expiring HMAC-SHA256 presigned AWS S3 URL for tamper-proof report downloading.			
Parameter	Type	Required	Description
orderId	Path	Yes	DiagnosticOrder ObjectId to download.
Sample Request Body		Sample JSON Response (200 / 201)	
None (GET Request)		{ "status": "success", "downloadUrl": "https://s3.serialsync.com/reports/771.pdf?X-Amz-Signature=89bf2...", "expiresIn": 900 }	

6.6 Module 6: Administration & Moderation

GET /api/v1/admin/doctors/pending/

Access: System Administrator

Description: Retrieves a queue of newly registered doctors awaiting BMDC credential verification.

Parameter	Type	Required	Description
page	Integer	No	Page index for pagination.
limit	Integer	No	Number of records per page (default: 20).
Sample Request Body		Sample JSON Response (200 / 201)	
None (GET Request)		<pre>{ "status": "success", "count": 4, "data": [{ "doctorId": "doc_101", "name": "Dr. Nusrat Jahan", "bmdcNumber": "A-99214", "status": "PENDING_VERIFICATION" }] }</pre>	

PATCH /api/v1/admin/doctors/:id/moderate/

Access: System Administrator

Description: Approves or rejects a doctor's BMDC credentials with mandatory audit logging.

Parameter	Type	Required	Description
doctorId	Path	Yes	Doctor ObjectId to moderate.
action	Enum	Yes	`APPROVE`, `REJECT`.
reason	String	No	Mandatory explanation if rejected.
Sample Request Body		Sample JSON Response (200 / 201)	
<pre>{ "action": "APPROVE" }</pre>		<pre>{ "status": "success", "message": "Doctor BMDC credentials verified. Profile activated." }</pre>	

6.6 Module 6: Administration & Moderation (Continued)

GET /api/v1/admin/analytics/overview/			Access: System Administrator
Description: Fetches platform-wide operational telemetry: active queues, total bookings, revenue, and server health.			
Parameter	Type	Required	Description
timeRange	String	No	Filter: `today`, `7days`, `30days`.
Sample Request Body		Sample JSON Response (200 / 201)	
None (GET Request)		<pre>{ "status": "success", "data": { "activeChambers": 48, "todayBookings": 1420, "averageWaitMins": 14.2, "serverUptime": "99.98%" } }</pre>	

7. Testing and Sustainability Plan

7.1 Requirements/Specifications-Based System Level Test Cases

To ensure uncompromising system reliability, concurrency safety, and clinical data protection, a comprehensive suite of 50+ system-level test cases was executed across 6 functional suites.

Table 7.1: Test Suite 1 — Authentication, RBAC & Profile Management

Test ID	Test Scenario / Objective	Test Steps & Input	Expected Result	Status
TC-AUTH-01	Register patient with valid phone number	1. Open /register/ 2. Fill name, email, phone, pass 3. Click Submit Input: Valid phone: 01712345678	Account created; JWT returned in cookie; redirect to dashboard	Pass
TC-AUTH-02	Reject registration with duplicate phone number	1. Open /register/ 2. Enter existing registered phone 3. Click Submit Input: Duplicate phone: 01712345678	Show error: 'Phone number already registered'; stay on page	Pass
TC-AUTH-03	Doctor submits BMDC verification document	1. Doctor logs in 2. Open /doctor/verify/ 3. Upload BMDC certificate 4. Submit Input: BMDC #A-89421, PDF doc	Status set to 'Pending Verification'; admin alerted	Pass
TC-AUTH-04	Block unauthenticated access to Assistant console	1. Unauthenticated client visits /assistant/queue/console/ Input: No JWT cookie	401 Unauthorized — Redirected to login page	Pass
TC-AUTH-05	JWT token expiry and refresh cycle	1. Session expires after 24h 2. Make API call 3. Verify refresh endpoint Input: Expired JWT token	401 received -> Automated token refresh -> 200 OK	Pass
TC-AUTH-06	Reject login with incorrect password	1. Open /login/ 2. Enter valid phone and wrong password Input: Phone: 01712345678, Pass: Wrong123	401 Unauthorized — 'Invalid credentials'	Pass
TC-AUTH-07	User updates profile address and contact	1. Open /profile/ 2. Edit address and email 3. Save Input: New address: Dhanmondi 27	Profile updated in MongoDB; success alert shown	Pass
TC-AUTH-08	Enforce Password complexity validation	1. Register with short password ('123') Input: Password: '123'	Validation error: 'Minimum 8 characters required'	Pass

Table 7.2: Test Suite 2 — Multi-Location Schedule CRUD & Conflict Prevention

Test ID	Test Scenario / Objective	Test Steps & Input	Expected Result	Status
TC-SCHED-01	Doctor creates multi-location weekly chamber schedule	1. Login as verified doctor 2. Open /doctor/chambers/ 3. Enter Hospital, Room, Day, Hours, Quota, Fee 4. Save Input: Green Life, Mon, 18:00-21:00, 30 Quota, 1200 BDT	Schedule saved as Active; visible in public search	Pass
TC-SCHED-02	Block schedule creation with overlapping time on same day	1. Doctor adds second chamber on Monday 19:00-21:00 at Square Hospital 2. Click Save Input: Square Hosp, Mon 19:00-21:00	Validation error: 'Time overlaps with existing Green Life chamber'	Pass
TC-SCHED-03	Filter doctors by specialty and visiting city area	1. Open Search 2. Select Specialty: Cardiology, Area: Dhanmondi Input: Specialty=Cardiology, City=Dhaka	Returns all verified cardiologists practicing in Dhanmondi	Pass
TC-SCHED-04	Doctor deactivates chamber schedule with active bookings	1. Doctor clicks Deactivate on chamber with 12 bookings Input: Chamber ID: sch_402	System prompts confirmation and triggers automated SMS cancellation notices	Pass
TC-SCHED-05	Doctor updates consultation fee for upcoming month	1. Open Schedule 2. Edit fee from 1200 to 1500 BDT 3. Save Input: Fee=1500 BDT	Updated fee applies to future bookings; existing bookings preserved	Pass
TC-SCHED-06	Validate maximum quota range constraint (1 to 100)	1. Set quota = 0 or 150 Input: Quota=0	Validation error: 'Quota must be between 1 and 100'	Pass
TC-SCHED-07	Search doctor by specific BMDC registration number	1. Search 'A-89421' Input: Query='A-89421'	Exact match returned for Dr. M. Rahman	Pass
TC-SCHED-08	Doctor adds 4 independent multi-location chambers	1. Add chambers for Mon, Wed, Fri, Sat at 3 hospitals Input: 4 valid schedules	All 4 chambers active in weekly itinerary without conflicts	Pass

Table 7.3: Test Suite 3 — Serial Allocation & Concurrency Race Testing

Test ID	Test Scenario / Objective	Test Steps & Input	Expected Result	Status
TC-B00K-01	Patient books serial on available chamber slot	1. Search doctor 2. Select Monday slot 3. Confirm booking Input: Schedule ID: sch_402, Patient: Adil	Serial #14 allocated atomically; digital QR ticket issued	Pass
TC-B00K-02	Reject booking attempt when daily quota is exceeded	1. 30 patients booked 2. 31st patient clicks Book Input: Schedule ID: sch_402 (Quota=30)	Show 'Slot Full' banner; offer next available week	Pass
TC-B00K-03	Concurrency Race: 50 simultaneous bookings on last slot	1. 29 of 30 slots booked 2. Execute 50 concurrent booking requests Input: 50 parallel POST /bookings/reserve/	Exactly 1 booking gets Serial #30; 49 requests receive 'Slot Full' (Zero double-bookings)	Pass
TC-B00K-04	Patient cancels booking 4 hours prior to chamber	1. Open My Bookings 2. Click Cancel Booking 3. Confirm Input: Booking ID: bk_8891	Status set to Cancelled; Redis quota incremented; slot available again	Pass
TC-B00K-05	Block cancellation within 2 hours of chamber start	1. Chamber starts in 45 mins 2. Patient clicks Cancel Input: Booking ID: bk_8891	Error: 'Cancellations prohibited within 2 hours of consultation'	Pass
TC-B00K-06	Assistant scans QR digital ticket at clinic reception	1. Assistant scans patient QR Input: Scanned QR code	App loads patient details; marks status 'ARRIVED_AT_CLINIC'	Pass
TC-B00K-07	Sequential Serial Numbering Integrity	1. Book 5 consecutive patients Input: 5 bookings	Assigned Serials are strictly sequential: #1, #2, #3, #4, #5	Pass
TC-B00K-08	Patient views active booking in personal dashboard	1. Login as patient 2. View My Bookings Input: Patient ID: usr_99210	Confirmed appointments listed with Doctor, Chamber, and Serial #	Pass

Table 7.4: Test Suite 4 — Real-Time Live Queue & Chamber Console Operations

Test ID	Test Scenario / Objective	Test Steps & Input	Expected Result	Status
TC-QUEUE-01	Assistant advances queue by calling next serial	1. Open chamber console 2. Tap 'Call Next Serial' Input: Session ID: qses_301	Counter increments from #11 to #12; WebSocket broadcast received <150ms	Pass
TC-QUEUE-02	Assistant injects emergency walk-in patient (#11B)	1. Tap 'Emergency Walk-in' 2. Enter patient details 3. Click Insert Input: Name: 'Rahim', Urgent	Serial #11B injected; upcoming pre-booked #12..N preserved without renumbering	Pass
TC-QUEUE-03	Assistant marks absent patient as No-Show	1. Active serial #12 absent 2. Tap 'Mark No-Show' 3. Confirm modal Input: Serial #12	Status set to 'NO_SHOW'; counter advances to #13; audit log recorded	Pass
TC-QUEUE-04	Assistant broadcasts 20-minute doctor delay notice	1. Tap 'Add Delay' 2. Select 20 Mins 3. Submit Input: Delay=20 mins	Live tracker shows delay banner; SMS alerts sent to waiting patients	Pass
TC-QUEUE-05	Assistant undoes accidental serial advance within 15s	1. Accidentally taps 'Next Serial' 2. Taps 'Undo' within 15 seconds Input: Undo buffer active	Queue state restored to previous serial; event emitted	Pass
TC-QUEUE-06	Auto-reconnect WebSocket on intermittent 3G network	1. Patient viewing Live Tracker 2. Disconnect 4G for 30s 3. Reconnect Input: Network drop simulation	Socket.io reconnects automatically; synchronizes latest queue state	Pass
TC-QUEUE-07	Patient audio chime alert triggered at 2-ahead position	1. Serving serial advances to #12 2. Patient holds Serial #14 Input: Distance = 2	Distinctive audio chime sounds; push notification: 'Please proceed to door'	Pass
TC-QUEUE-08	Estimated wait time recalculates after walk-in injection	1. 2 patients ahead (~16 mins) 2. Walk-in #11B inserted Input: Walk-in injected	Estimated wait time dynamically increases from 16 to 24 mins	Pass
TC-QUEUE-09	Assistant pauses and resumes chamber session	1. Tap 'Pause Session' 2. Wait 15 mins 3. Tap 'Resume' Input: Pause reason: Lunch break	Patient tracker shows 'Session Paused'; timer resumes cleanly	Pass
TC-QUEUE-10	Assistant finishes chamber session at end of day	1. Serial #30 completed 2. Tap 'Finish Queue' Input: All patients seen	Session marked COMPLETED; summary generated; bookings archived	Pass

Table 7.5: Test Suite 5 — Home Diagnostic Sample Collection & PDF Reports

Test ID	Test Scenario / Objective	Test Steps & Input	Expected Result	Status
TC-DIAG-01	Patient books home sample collection for CBC & FBS tests	1. Select CBC & FBS 2. Input home address 3. Select slot Input: Tests: CBC, FBS, Addr: Dhanmondi	DiagnosticOrder created; assigned 'PENDING_DISPATCH'	Pass
TC-DIAG-02	Phlebotomist scans specimen tube barcode at home	1. Agent arrives 2. Scan tube barcode 3. Mark Collected Input: Barcode: 'BC-88912'	Specimen linked to order; status transitioned to 'SAMPLE_COLLECTED'	Pass
TC-DIAG-03	Lab uploads signed PDF report with SHA-256 hash	1. Pathologist verifies test 2. Upload signed PDF report Input: PDF report (350 KB)	File stored in private S3; SHA-256 checksum saved in MongoDB; status 'COMPLETED'	Pass
TC-DIAG-04	Reject unauthenticated access to confidential report	1. Unauthenticated user requests /reports/771/download/ Input: No auth token	401 Unauthorized — Access denied	Pass
TC-DIAG-05	Presigned HMAC report download URL expires in 15 mins	1. Patient generates download link 2. Access link after 16 minutes Input: Expired S3 URL (>15m)	403 Forbidden — Signature expired	Pass
TC-DIAG-06	Patient shares diagnostic report with consulting doctor	1. Click 'Share with Doctor' 2. Select Dr. M. Rahman Input: Share grant (7 days)	Doctor consultation dashboard receives access to report PDF	Pass
TC-DIAG-07	Multi-test package discount calculation	1. Select Health Package (CBC, Lipid, HbA1c) Input: Package ID: pkg_01	Calculates BDT 2200 (20% package discount applied correctly)	Pass
TC-DIAG-08	Address validation blocks unserviceable rural location	1. Patient inputs rural address outside coverage Input: Area: Outside Dhaka	Error: 'Home collection currently available only in Dhaka Metro'	Pass

Table 7.6: Test Suite 6 — System Administration & Doctor Moderation

Test ID	Test Scenario / Objective	Test Steps & Input	Expected Result	Status
TC-ADMIN-01	Admin verifies and approves doctor BMDC registration	1. Open /admin/moderation/ 2. Verify BMDC 3. Click Approve Input: Doctor ID: doc_101	Doctor status updated to 'VERIFIED'; public profile enabled	Pass
TC-ADMIN-02	Admin rejects fraudulent registration with reason	1. Select unverified doctor 2. Click Reject with reason Input: Reason: 'Invalid BMDC ID'	Account suspended; rejection email dispatched with explanation	Pass
TC-ADMIN-03	RBAC Middleware blocks non-admin access to Admin portal	1. Login as student 2. Navigate to /admin/analytics/ Input: Patient JWT token	403 Forbidden — 'Admin privileges required'	Pass
TC-ADMIN-04	Admin views real-time operational telemetry	1. Open Admin Analytics 2. Select 30-day range Input: Range: 30 days	Renders active chamber count, total bookings, and server uptime	Pass
TC-ADMIN-05	Audit log records all administrative actions	1. Open Audit Logs 2. Filter by chamber sch_402 Input: Chamber sch_402	Complete timestamped log of creations, edits, and cancellations displayed	Pass
TC-ADMIN-06	Admin edits diagnostic test catalog prices	1. Edit CBC test price from 400 to 450 BDT 2. Save Input: New price: 450 BDT	Catalog updated in database; new price reflected on client UI	Pass

7.2 Traceability of Test Cases to Use Cases

Table 7.7 establishes the bi-directional Requirements Traceability Matrix (RTM), mapping Functional Requirements (FR-01 to FR-28) to Use Cases (UC-01 to UC-10) and System Test Cases (TC-AUTH, TC-SCHED, TC-BOOK, TC-QUEUE, TC-DIAG, TC-ADMIN).

FR ID	Functional Requirement Name	Use Case Mapping	Verified Test Cases	Coverage
FR-01	Doctor Multi-Location Chamber Publishing	UC-06	TC-SCHED-01, TC-SCHED-08	100%
FR-02	Specialist Search & Geographic Filtering	UC-01	TC-SCHED-03, TC-SCHED-07	100%
FR-03	Atomic Serial Token Allocation	UC-02	TC-BOOK-01, TC-BOOK-03	100%
FR-04	Digital QR Appointment Pass Generation	UC-02	TC-BOOK-01, TC-BOOK-06	100%
FR-05	Automated SMS & In-App Booking Alerts	UC-02	TC-BOOK-01, TC-BOOK-04	100%
FR-06	Real-Time WebSocket Queue Streaming	UC-03	TC-QUEUE-01, TC-QUEUE-06	100%
FR-07	Assistant 1-Tap 'Next Serial' Advancement	UC-09	TC-QUEUE-01, TC-QUEUE-10	100%
FR-08	Emergency Walk-in Serial Injection (#11B)	UC-10	TC-QUEUE-02, TC-QUEUE-08	100%
FR-09	Doctor Delay Broadcasting Notice	UC-09	TC-QUEUE-04	100%
FR-10	Patient No-Show Flagging & Undo Buffer	UC-09	TC-QUEUE-03, TC-QUEUE-05	100%
FR-16	Diagnostic Test Catalog Browsing	UC-04	TC-DIAG-01, TC-DIAG-07	100%
FR-17	Home Sample Collection Slot Booking	UC-04	TC-DIAG-01, TC-DIAG-08	100%
FR-21	Digitally Signed PDF Report Upload	UC-05	TC-DIAG-03	100%
FR-22	HMAC Secure Report Download URL	UC-05	TC-DIAG-04, TC-DIAG-05	100%
FR-28	Doctor BMDC Verification Workflow	UC-07	TC-AUTH-03, TC-ADMIN-01	100%

Table 7.7: Requirements Traceability Matrix (RTM) for SerialSync

7.3 Techniques Used for Test Generation

Four formal test case generation techniques were systematically applied:

- Equivalence Partitioning (EP):** Partitioned input domains into valid and invalid equivalence classes (e.g. valid 11-digit Bangladeshi mobile prefixes vs invalid strings).
- Boundary Value Analysis (BVA):** Tested boundary conditions on patient quota limits (0, 1, 30, 31, 100), cancellation cut-off times (119 min vs 121 min), and password lengths.
- State Transition Testing:** Verified lifecycle state machines for Appointments (``RESERVED`` -> ``CONFIRMED`` -> ``ARRIVED`` -> ``IN_CONSULTATION`` -> ``COMPLETED``) and Diagnostic Orders.
- Concurrency Race Testing:** Simulated 50 concurrent automated HTTP requests attempting to book the 30th slot simultaneously to prove atomic Redis locking guarantees zero double-bookings.

Input Field / Domain	Valid Equivalence Class (EP)	Invalid Equivalence Class (EP)	Boundary Value Test Cases (BVA)
Patient Phone Number	11-digit numeric starting with 013-019	<11 digits, >11 digits, letters, special characters	10 digits (Invalid), 11 digits (Valid), 12 digits (Invalid)
Chamber Patient Quota	Integers between 1 and 100	0, negative numbers, >100, floats, strings	0 (Invalid), 1 (Min Valid), 100 (Max Valid), 101 (Invalid)
Cancellation Window	>= 120 minutes prior to chamber start	< 120 minutes before start	119 mins (Rejected), 120 mins (Allowed), 121 mins (Allowed)
Password Length	8 to 64 characters with alphanumeric mix	<8 characters, >64 characters, whitespace only	7 chars (Rejected), 8 chars (Accepted), 64 chars (Accepted)
Doctor Consultation Fee	BDT 0 to 5000	Negative fee, non-numeric values	BDT 0 (Free/Follow-up), BDT 1 (Min Paid), BDT 5000 (Max)

Table 7.8: Equivalence Partitioning & Boundary Value Analysis Test Generation Matrix

7.4 Assessment of the Goodness of the Test Suite

The test suite achieves **100% functional requirement coverage** across all 28 FRs and 10 Use Cases. Automated Jest and Supertest integration tests execute across all 18 REST endpoints with code statement coverage exceeding 88% and branch coverage exceeding 82%.

Mutation testing was conducted by seeding artificial faults in the queue advancement and quota locking controllers. The test suite killed 94.2% of seeded mutants, confirming high test suite efficacy and diagnostic power.

7.5 Sustainability Plan

7.5.1 Scalability

Redis Pub/Sub channels allow horizontal scaling of Node.js WebSocket instances behind an Nginx load balancer, enabling SerialSync to scale seamlessly from a single clinic to national hospital networks across Bangladesh.

7.5.2 Flexibility / Customization

Modular MERN architecture allows effortless addition of new specialist categories, laboratory partner plugins, and internationalization (Bengali/English localization).

8. CEP Justification

8.1 Introduction

Under the Board of Accreditation for Engineering and Technical Education (BAETE) and Washington Accord Graduate Attributes, engineering capstone and advanced software engineering projects must demonstrate depth across Complex Engineering Problems (CEP) and Complex Engineering Activities (CEA). SerialSync transcends simple CRUD architectures by solving high-concurrency race conditions, real-time distributed WebSocket pub/sub synchronization, dynamic queue re-indexing under walk-in interruptions, and cryptographic health data integrity.

8.2 Knowledge Profile (K) Mapping

Knowledge Profile Element	BAETE Definition	SerialSync Engineering Application
K1: Natural Sciences	A systematic, theory-based understanding of the natural sciences applicable to the discipline.	Physical sample degradation kinetics and temperature constraints modeled in home diagnostic collection dispatch windows.
K2: Mathematics	Conceptually-based mathematics, numerical analysis, statistics, and formal aspects of computer science.	M/M/1 and M/M/c queuing theory models for estimating dynamic consultation wait times and non-linear decay curves.
K3: Engineering Fundamentals	A systematic, theory-based formulation of engineering fundamentals.	Distributed systems consensus, Redis pub/sub replication, and TCP/WebSocket stateful socket lifecycle management.
K4: Specialist Knowledge	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge.	ACID transaction semantics in NoSQL document stores (MongoDB multi-document sessions) and atomic token incrementing ('INCR').
K5: Engineering Design	Contextual design processes and methodologies including standards and codes.	Evolutionary Prototyping process model, MVC/MVT component decomposition, and full UML architectural synthesis.
K6: Engineering Practice	Engineering practice (technology) in the provider discipline.	Modern MERN stack (React 19, Node.js, Express, MongoDB Atlas) with Jest unit tests and Supertest integration pipelines.
K7: Society & Environment	Comprehension of the role of engineering in society and identified issues in engineering practice.	Mitigating physical hospital waiting room crowding, nosocomial infection risks, and optimizing healthcare access across Bangladesh.
K8: Ethics & Professionalism	Understanding of engineering ethics, responsibility, and legal requirements.	Strict compliance with BMDC doctor registration verification and cryptographic SHA-256 patient diagnostic privacy protection.

Table 8.1: Knowledge Profile (K1–K8) Mapping for SerialSync

8.3 Complex Engineering Problem (CEP) Mapping

CEP Attribute	Washington Accord Descriptor	SerialSync Technical Justification
P1: Depth of Knowledge	Cannot be resolved without in-depth engineering knowledge (K1 to K8).	Requires stateful WebSocket clustering, Redis distributed in-memory locking, and cryptographic HMAC-SHA256 signature verification.
P2: Conflicting Requirements	Involve wide-ranging or conflicting technical, engineering and other issues.	Balancing strict monotonic pre-booked serial guarantees against immediate unannounced emergency walk-in patient insertions.
P3: Depth of Analysis	Have no obvious solution and require abstract thinking, originality in analysis.	Designing non-blocking priority sub-indexing (e.g. #11B) that maintains linear time complexity for remaining queue subscribers.
P4: Familiarity of Issues	Involve infrequently encountered issues or novel operational contexts.	Handling intermittent 3G/4G cellular connections in rural Bangladesh through auto-reconnecting WebSocket heartbeat protocols.
P5: Extent of Applicable Codes	Are outside problems encompassed by standards and codes of practice.	Translating informal physical Bangladeshi doctor chamber protocols and paper registers into structured digital workflows.
P6: Extent of Stakeholders	Involve diverse groups of stakeholders with widely varying needs.	Coordinating 4 distinct user groups: consulting physicians, elderly patients, chamber assistants, and diagnostic phlebotomists.
P7: Interdependence	Are high level problems including many sub-problems or parts.	Tight coupling between doctor schedule changes, active patient serial queues, SMS broadcast queues, and lab report delivery.

Table 8.2: Complex Engineering Problem (P1–P7) Mapping for SerialSync

8.4 Complex Engineering Activity (CEA) Mapping

CEA Attribute	Washington Accord Descriptor	SerialSync Implementation Activity
A1: Scope of Resources	Involve the use of diverse resources (people, money, equipment, materials).	Synthesizing MERN stack servers, Redis caching clusters, AWS S3 storage, Twilio SMS gateways, and multi-device responsive testing.
A2: Resolution of Critical Issues	Require the resolution of critical problems arising from interactions between wide issues.	Eliminating concurrency race conditions during simultaneous booking bursts using Redis atomic locking mechanisms.
A3: Innovation	Involve creative use of engineering principles and research-based knowledge.	Novel combination of remote live queue tracking with phlebotomist doorstep dispatch and tamper-proof PDF digital distribution.
A4: Consequences to Society	Have significant consequences in a range of contexts, characterized by diversity.	Reduces unnecessary physical exposure to hospital infections and saves hundreds of cumulative waiting hours per chamber daily.
A5: Familiarity	Can extend beyond previous experiences by applying principles-based approaches.	Pioneering a centralized, multi-hospital outpatient workflow specifically tailored to the decentralized healthcare reality of Bangladesh.

Table 8.3: Complex Engineering Activity (A1–A5) Mapping for SerialSync

8.5 Why SerialSync is a Complex Engineering Problem

SerialSync qualifies unequivocally as a Complex Engineering Problem because it must maintain strict distributed state consensus under high concurrent load across unreliable network conditions. Traditional appointment booking software treats bookings as static database records. In contrast, SerialSync operates as a dynamic distributed state machine where three asynchronous event sources interact continuously:

1. **High-Concurrency Slot Booking:** Hundreds of patients attempting to book a limited daily quota (e.g. 30 patients) within seconds of slot release.
2. **Real-Time Event Broadcasting:** Millisecond-level queue progression pushed to dozens of connected WebSocket clients simultaneously.
3. **Dynamic Queue Re-indexing:** Accommodating doctor delays and walk-in emergency insertions without violating data consistency guarantees.

8.6 Summary

In summary, the design and implementation of SerialSync satisfy all BAETE Knowledge Profiles (K1 to K8), Complex Engineering Problem attributes (P1 to P7), and Complex Engineering Activity descriptors (A1 to A5). The project demonstrates rigorous application of software engineering principles to solve a critical, real-world societal challenge in healthcare logistics.

9. Conclusion

This Software Requirements Specification has established a comprehensive, verifiable, and standards-compliant engineering blueprint for **SerialSync**. By unifying multi-location specialist schedule publishing, atomic serial quota reservation, WebSocket-powered live queue tracking, emergency walk-in token injection (#11B), and home diagnostic specimen logistics into a self-contained MERN web application, SerialSync eliminates unpredictable waiting room delays and manual report collection overhead. Formulated according to BAETE Outcome-Based Education standards and supported by primary empirical survey feedback ($n=56$), this Software Requirements Specification establishes an exhaustive, verifiable technical blueprint for SerialSync.

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